

Affect is prominent in word meaning

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Affect-First Hypothesis

Wundt 1907; Zajonc 1980, 2000; Murphy & Zajonc 1993; LeDoux 1996; Stapel, Koomen & Ruys 2002

"You d*mb B*TCH!"

"Eat me."



"You d*mb B*TCH!"

"Hwhyyyyy?"



Roadmap

A. Prelude

B. AFH, Evidence, Psych., Ling.

C. Conceptual Domains

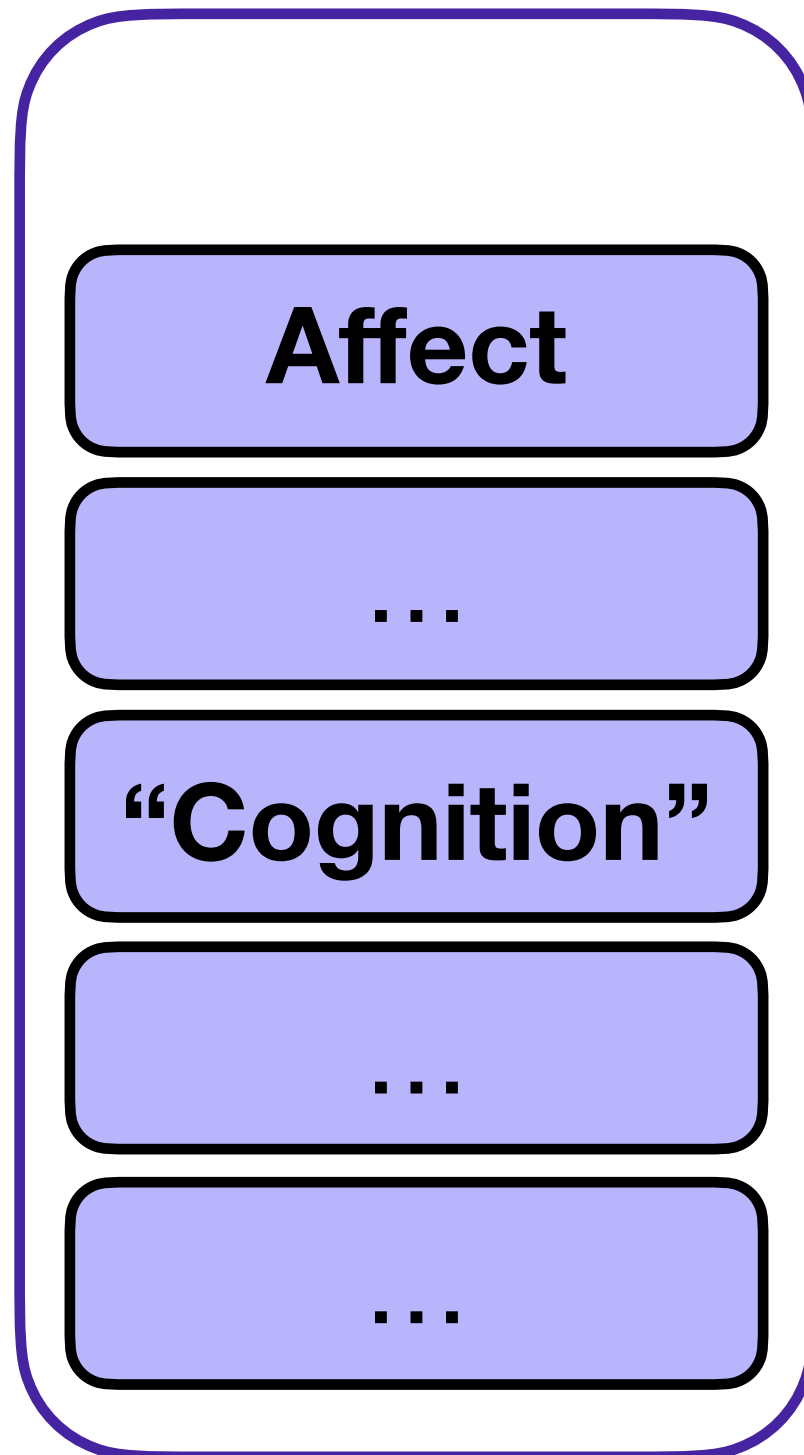
D. Experiments 1-7: Categorization

E. Handling (some) objections

F. Experiments 8-14: Similarity

G. Discussion/Conclusion

AFH ~ prominence



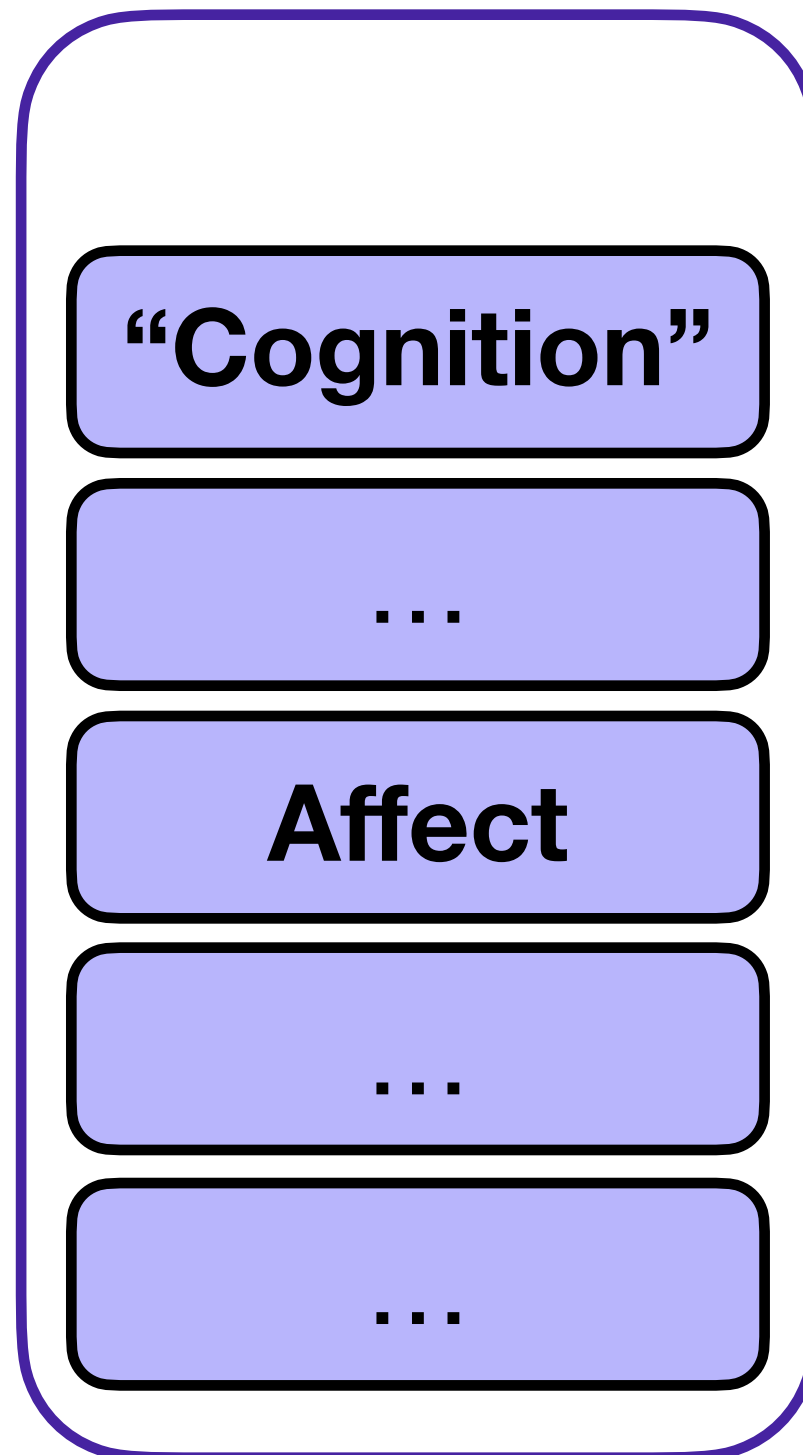
Evidence for AFH

Participants respond faster + early ERP responses to Emo than neutral words (Costa et al 2009; Vigliocco et al 2011; Scott et al 2012)

Mere Exposure Effect (Zajonc 1968) and Affective priming (Fazio 2001)

Valence congruity facilitates semantic evaluation in patients with semantic dementia (Souter et al. 2023)

The Alternative



Against...

**Object recognition (visual scene or facial)
required for affect effects** (Lazarus 1984; Rolls 1999;
Storbeck, Robinson & McCour 2006; Storbeck & Clore 2007; Calvo
& Nummenmaa 2007; Batty, Cave, & Pauli, 2005; Fox et al., 2000;
Horstmann & Bauland, 2006; Nummenmaa, Hyönä & Calvo 2010)

Oversimplification?

Kober et al.

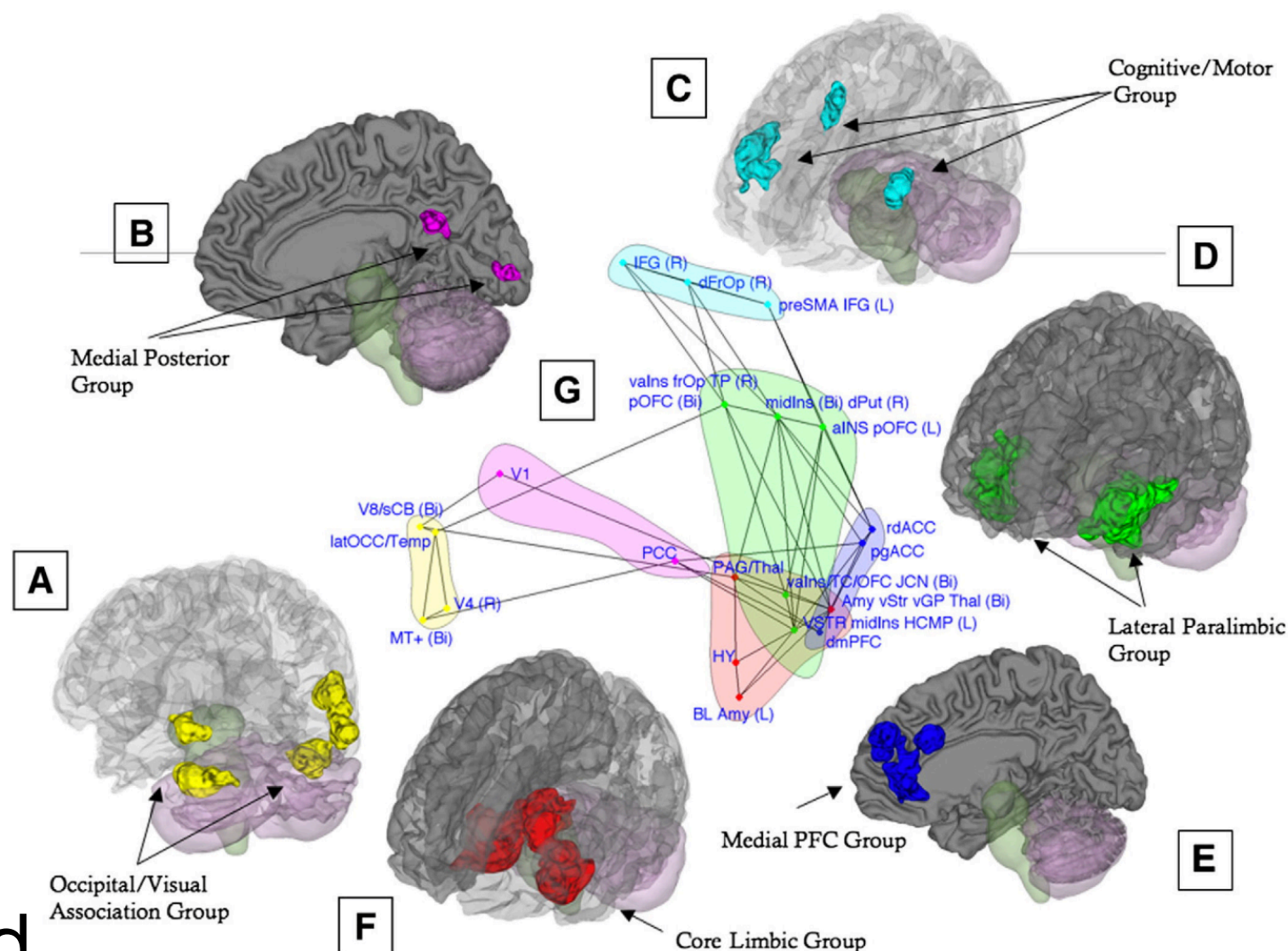
Page 49

AFH assumes independence of affect and “cognition”. Is it correct? (Souter et al ; Korber 2008)

Korber et al found 6 circuits of coactivation

Affect vs. “*cognitive*” content: it’s complicated

(Winter 2022; Hinojosa et al 2019; Hinojosa et al 2023)



Interacting variables

- Word frequency
- Word length
- Native/non-native
- Gender differences
- Morphosyntax
- Concreteness
- Imageability
- Age of acquisition
- Familiarity
- Contextual availability
- Semantic relatedness

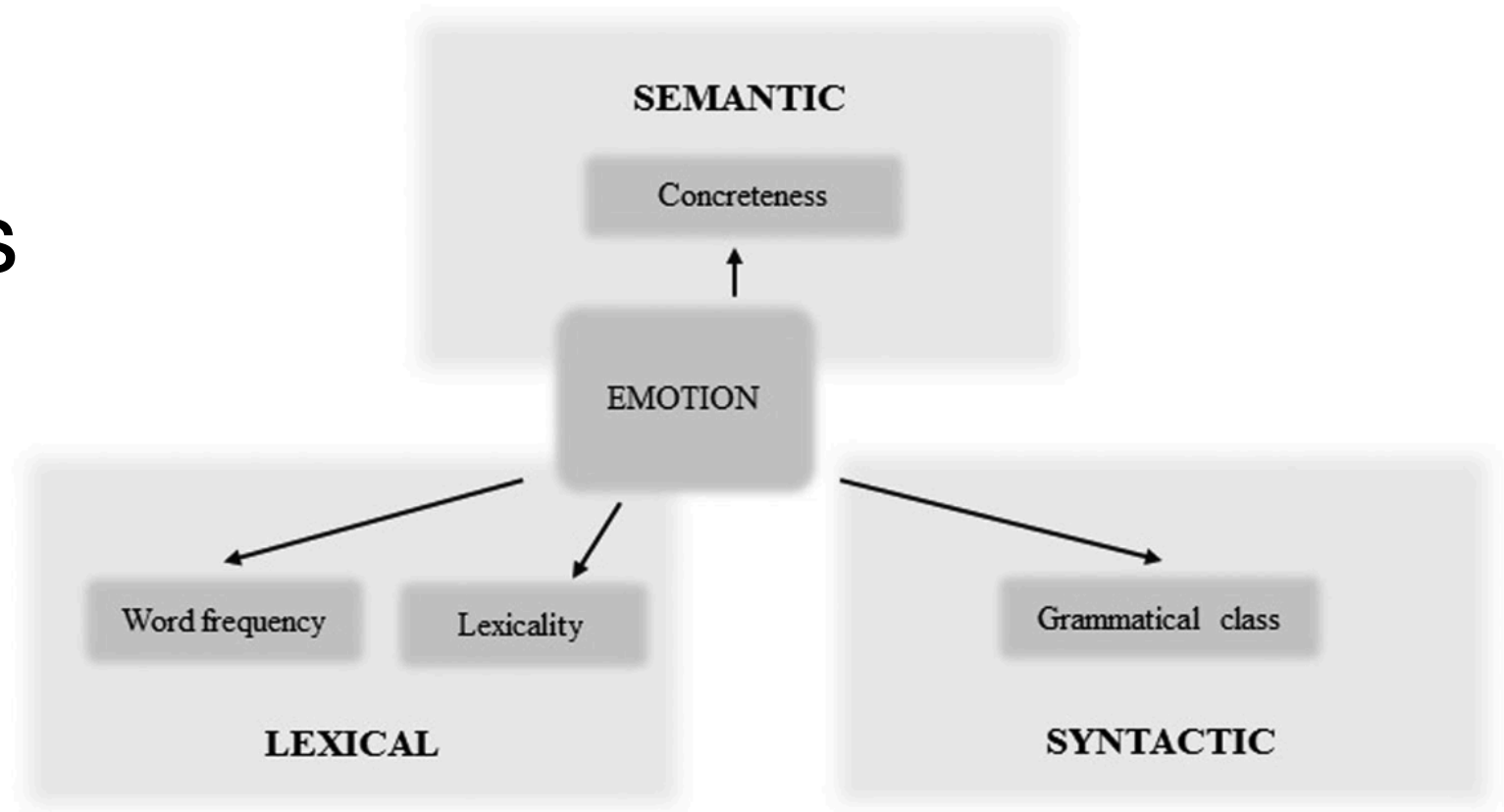


Figure 2. Relations between emotional features and different lexical, semantic and syntactic features at the word level.

Hinojosa, Moreno & Ferré (2019)

Affect in context

Task-dependent & sensitive to psycholinguistic variables

(Hinojosa et al 2014, Kaltwasser et al 2013; Hinojosa et al 2019)

Appear when words describe people or are in communicative contexts

(Rohr & Abdel Rahman 2018; Rohr & Wentura 2022)

Direct listener attention to speaker intentions

(Rodrigues Ronderos & Domaneschi 2023)

Types of Evidence for AFH

Between-stimulus:

- Emo vs. neutral stimuli

Problem/Challenge:

1. Controlling for irrelevant non-affect content differences
2. Finding commensurate non-affect dimension

Within-stimulus:

- Affective vs. Conceptual dimensions of a single stimulus

Problem/Challenge:

2. Finding commensurate non-affect dimension

Affect & Words: Psych.

Affect is a part of all concepts

(Osgood et al 1957; Lambon Ralph et al 2017; Yap & Show 2017)

evaluation, potency, activity

(Osgood et al. 1957)

...

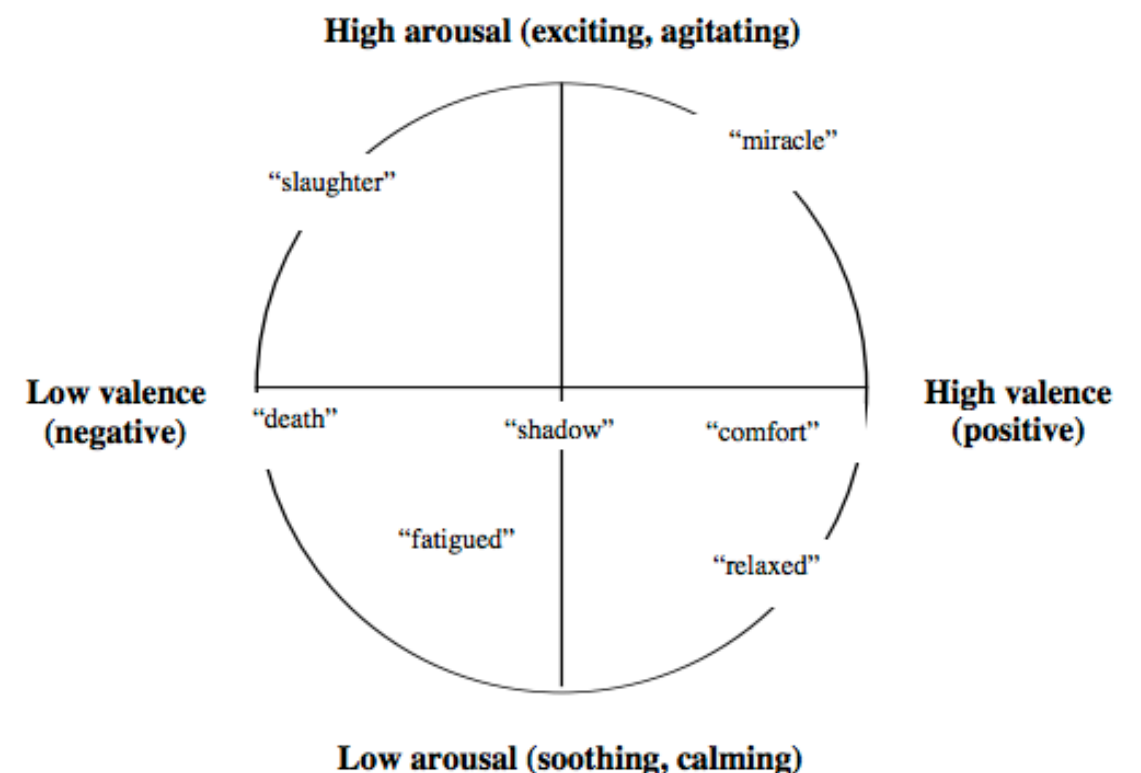
pleasure, arousal, dominance

(Mehrabian et Russell 1974)

...

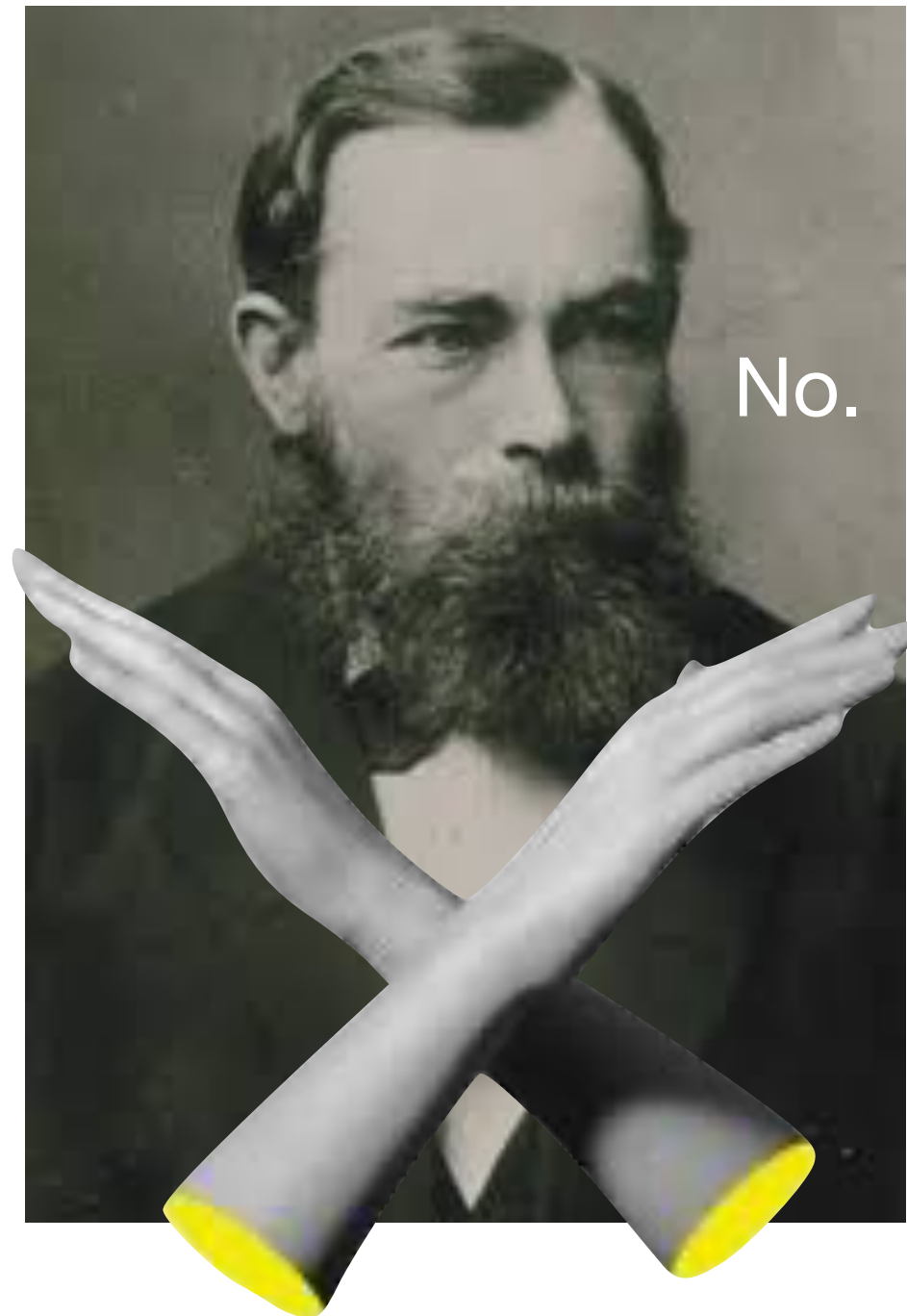
valence, arousal, dominance

(Bradley & Lang 1999; Warriner et al. 2013; Mohammad 2018)



the positive or
negative evaluation

Affect & Words: Ling.



Persistent puzzles to the Fregean program / meaning referentialism

Strawson (1950) “on referring”, Perry (1979) “Essential indexical”, Lewis (1979) “attitudes de dicto and de se”, Davidson (1986) “derangement of epitaphs”, Lewis (1996) “elusive knowledge”, Kaplan (1999) on “oops and ouch” & Kratzer’s discussion of it... everything in the pragmatic wastebasket

From ~2005 on explosion of work on expressives, slurs, evaluative, and other non-truth-conditional aspects of meaning

Potts (2006), McCready 2014, Gutzmann (2015), Cepollaro (2020), Jeshion (2021)...Stojanovic (2007), Stojanovic (2012)

Is Affect *REALLY* first?

Affect

Foil 1

Foil 2

Foil 3

Foil 4

Explicit (Exps 1-4)

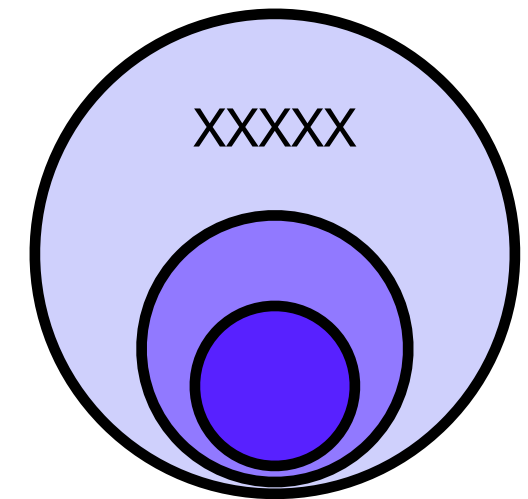
Is valence **categorization** faster?

Implicit (Exps 5-8)

Is valence more prominent for judging **word similarity**?

Conceptual foils

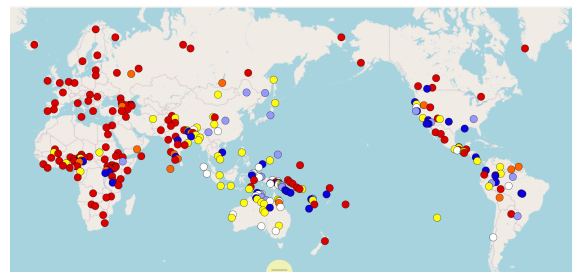
1. Concrete vs. abstract
(nouns, verbs, adjs)



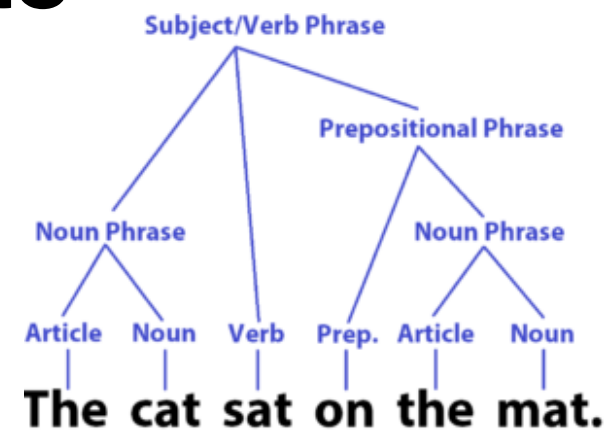
2. Physical vs. psych./social
(verbs, adjs)



3. Animate vs. inanimate
(nouns)

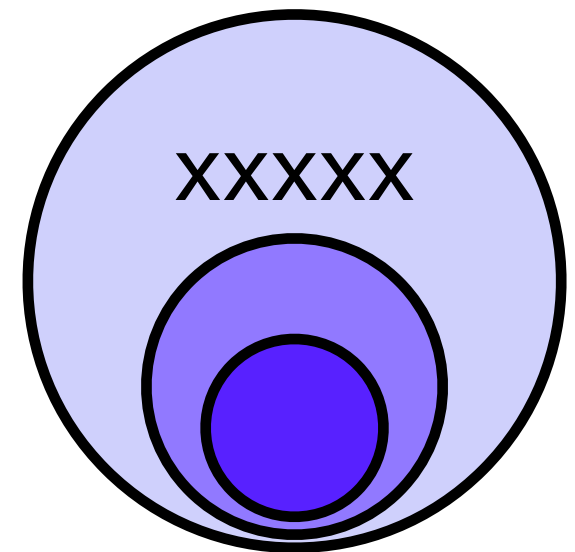


4. Syntax
(noun vs. verb)



1. Concrete/Abstract

Long-established as core dimension of concept meaning & interacts with affect



(Paivio 1971; James 1975; Rudin 1980; Paivio 1986; Kroll et al 1986; Schwanenflugel et al 1988; Binder et al 2005; Palazova et al 2003; Herbert et al; Kansuke & Kotz 2007; Koustas et al 2011; Meteyard et al 2012; Vigliocco et al 2014; Borghi et al 2017, 2018, 2019; Ponari et al, 2018; Barsalou et al 2018; Pauligk et al 2019; Hinojosa et al 2019; Lund et al 2019; Kim et al 2020; Winter 2022, ...)

Valence x Concreteness

Relationship Status: HS sweethearts but it's complicated

Emotion grounds abstract concepts: valence aids abstract word learning (Ponari et al, 2018, Lund et al 2019; Kim et al 2020), and activates similar brain areas (Vigliocco et al 2014)

“The inverted U”: the more abstract, the more valenced (Kousta et al 2011)

- **Yes But!** Winter (2020): depends on *particular* rating measures and *particular* statistical analyses matter

But Concrete words elicit valence contrasts earlier than abstract (Palazova et al 2003)

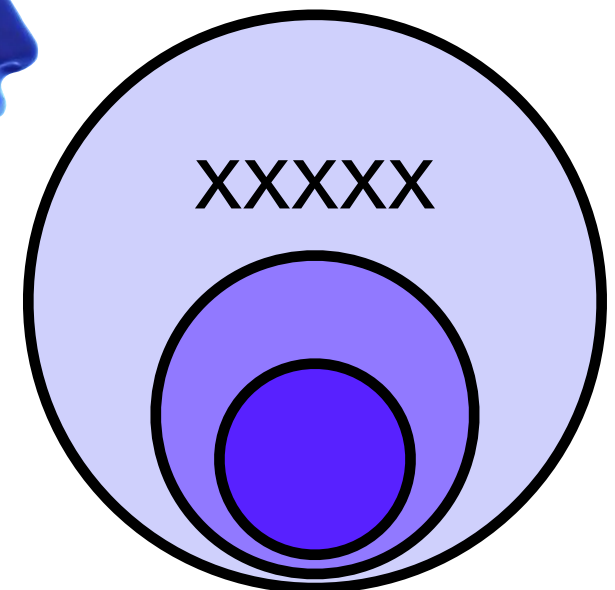
- Modulation by task demand (Kansuke & Kotz 2007), other features like frequency (Hinojosa et al 2019)

But ‘Concrete/Abstract’ too simple (Löhr 2021, Strike Levers et al 2021, Connell et al 2012; Pollock 2018; Juhasz et al 2013) ~ Lancaster sensorimotor norms with 11 dimensions! (Lynott et al, 2019)

2. Physical v. Psych./Social

Infants are attuned to many differences between physical vs. psychological/social

- Social agents create order but physical objects do not (Newman et al. 2010)
- Proximal causation for physical objects but not social agents / distal causation for social agents but not physical objects (Leslie & Keeble 1987, Spelke et al. 1995)
- Objects not capable of self-propelled motion but social agents are (Saxe et al. 2005)

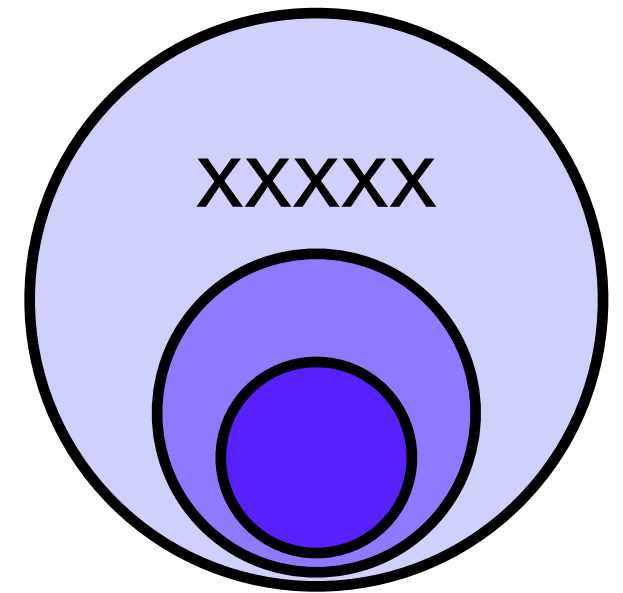


Remains important in adulthood: ascribing causation (Strickland et al. 2017), and iconicity and category boundaries (Kuhn et al. 2021)

3. Animacy

Developmentally/Cognitively Special

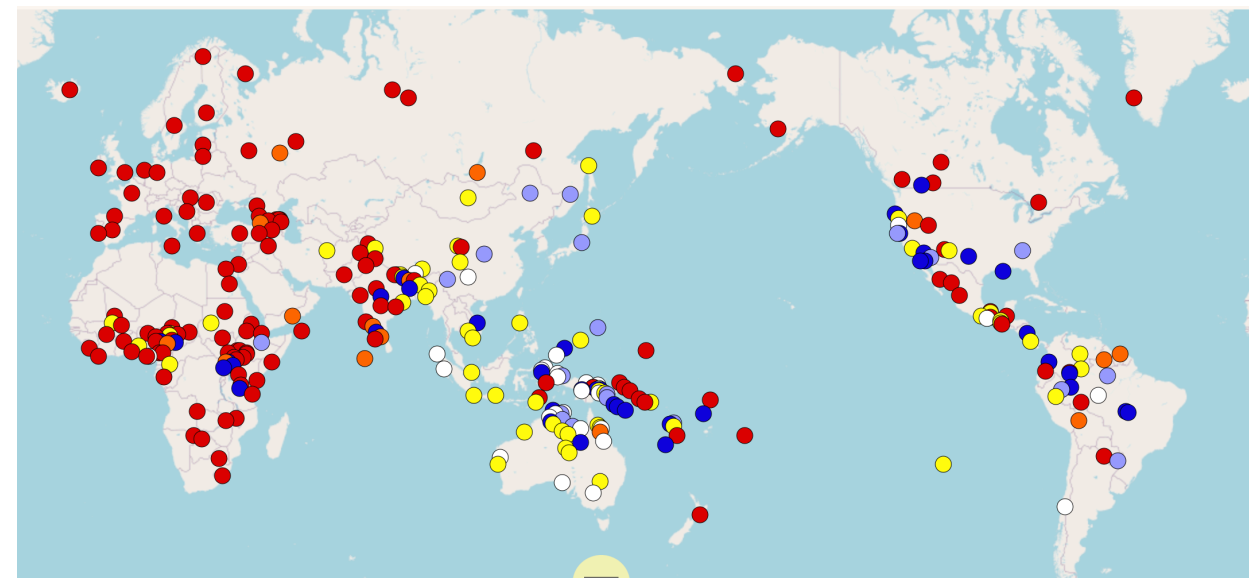
- Early (Rakison & Poulin-Dubois 2001), distinct neurophysiology (Caramazza & Shelton 1998)
- Causation (Spelke, Phillips, & Woodward 1995), word learning (Childers & Echols 2004), mindreading (Baron-Cohen 1995), biological processes (Carey 1985)



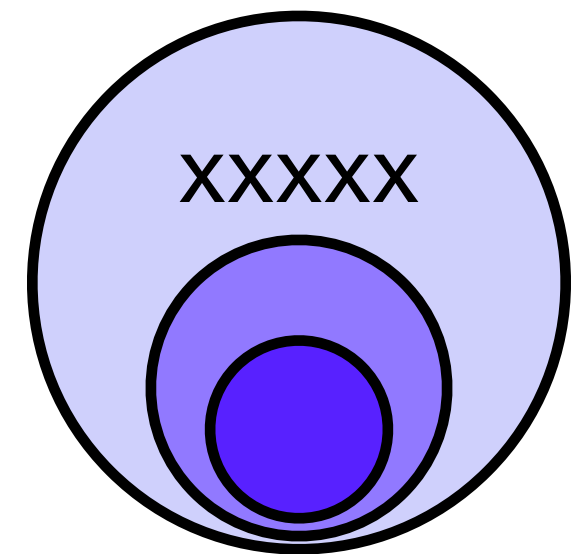
Linguistically Special

- Most if not all languages show grammatical reflexes of animacy (Vihman & Nelson 2019)

☐	• v	No nominal plural	28
☐	• v	Only human nouns, optional	20
☐	• v	Only human nouns, obligatory	40
☐	• v	All nouns, always optional	55
☐	• v	All nouns, optional in inanimates	15
☐	• v	All nouns, always obligatory	133



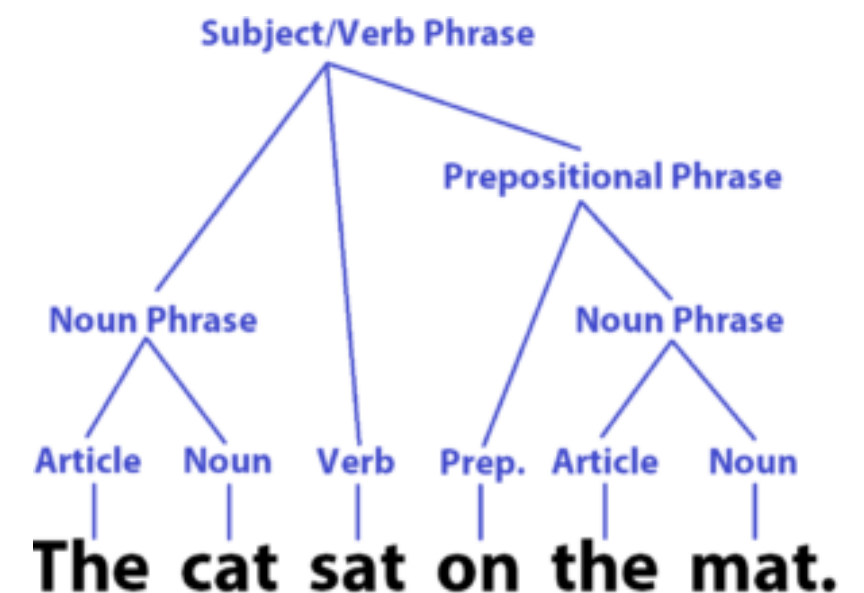
4. Syntax



Underlies combinatorial linguistic cognition

Syntactic information argued to be retrieved early on lexical access

(e.g., Strijkers & Costa 2011)



Roadmap

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D. Experiments 1-7: Categorization

E. Handling (some) objections

F. Experiments 8-14: Similarity

G. Discussion/Conclusion

GENOCIDE

F

positive

J

negative

GENOCIDE

F

abstract

J

concrete

Speeded categorization

2 x 1 within-subjects

blocked design (counterbalanced)

- Task (**Valence**, **Conceptual**)
- Items presented 3x per block (randomized)
- DM: key board press F or J (counterbalanced)

7 exps

270 Native American English speakers on Prolific, £9/hr, ~15 min median time

Stimuli

40 words in each exp, within lexical category

10 per 4 possible feature combination

Feature values came from corpus or pre-test norming

Stimuli from corpus/norms

- **Valence**: Warriner et al 2013
- **Conc. vs. Abs.**: Brysbaert et al 2013
- **Phys. vs. Psych./Soc.**: 2 pre-test norming studies conducted
- **Animacy**: Wordnet synsets with 'living organism' or 'alive' as superset (then hand-selected)
- **Syntax**: unambiguous nouns vs. verbs

RT Analysis

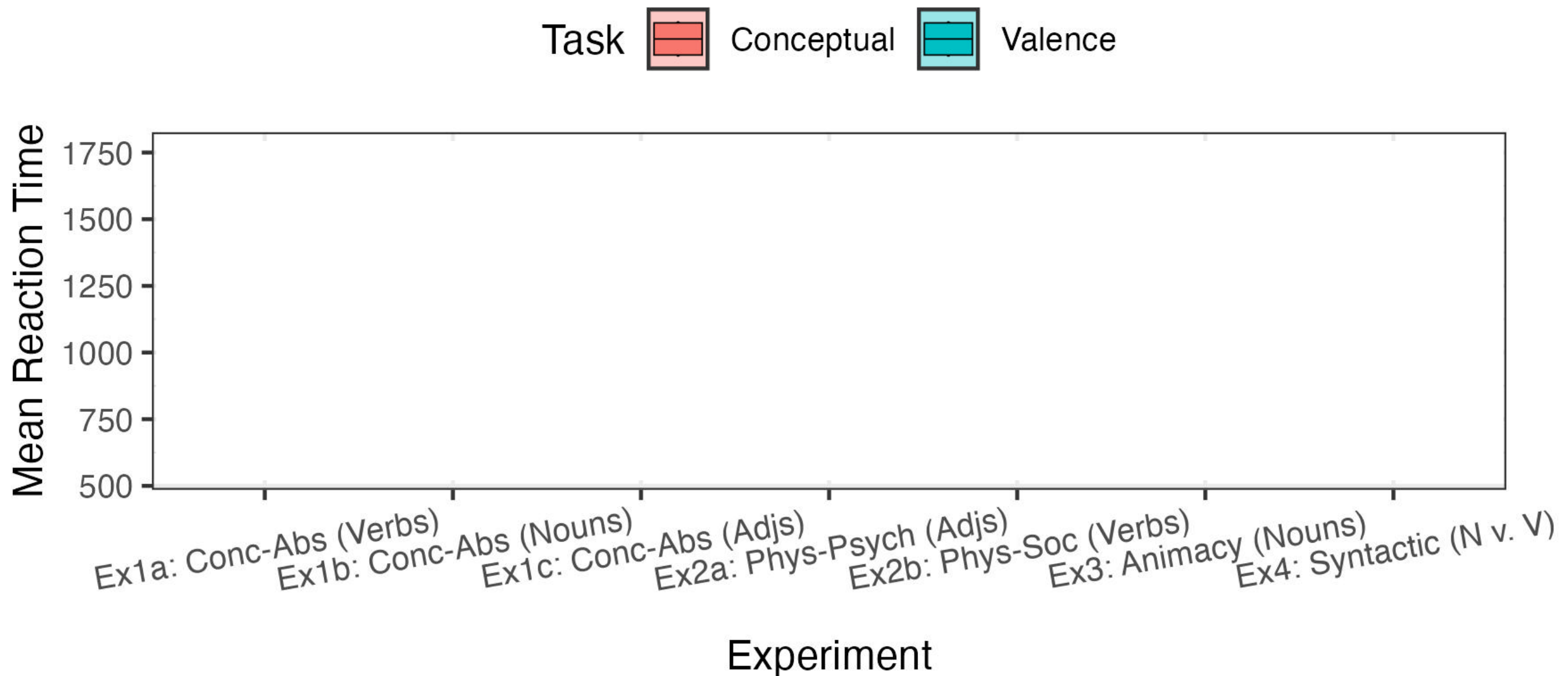
- Inaccurate participants removed ($\text{Acc} < 75\%$)
- Only Accurate trials included
- Excluded trials where $\text{LogRT} > 3 \times \text{IQR}$

Range of data excluded across exps: ~10-15%

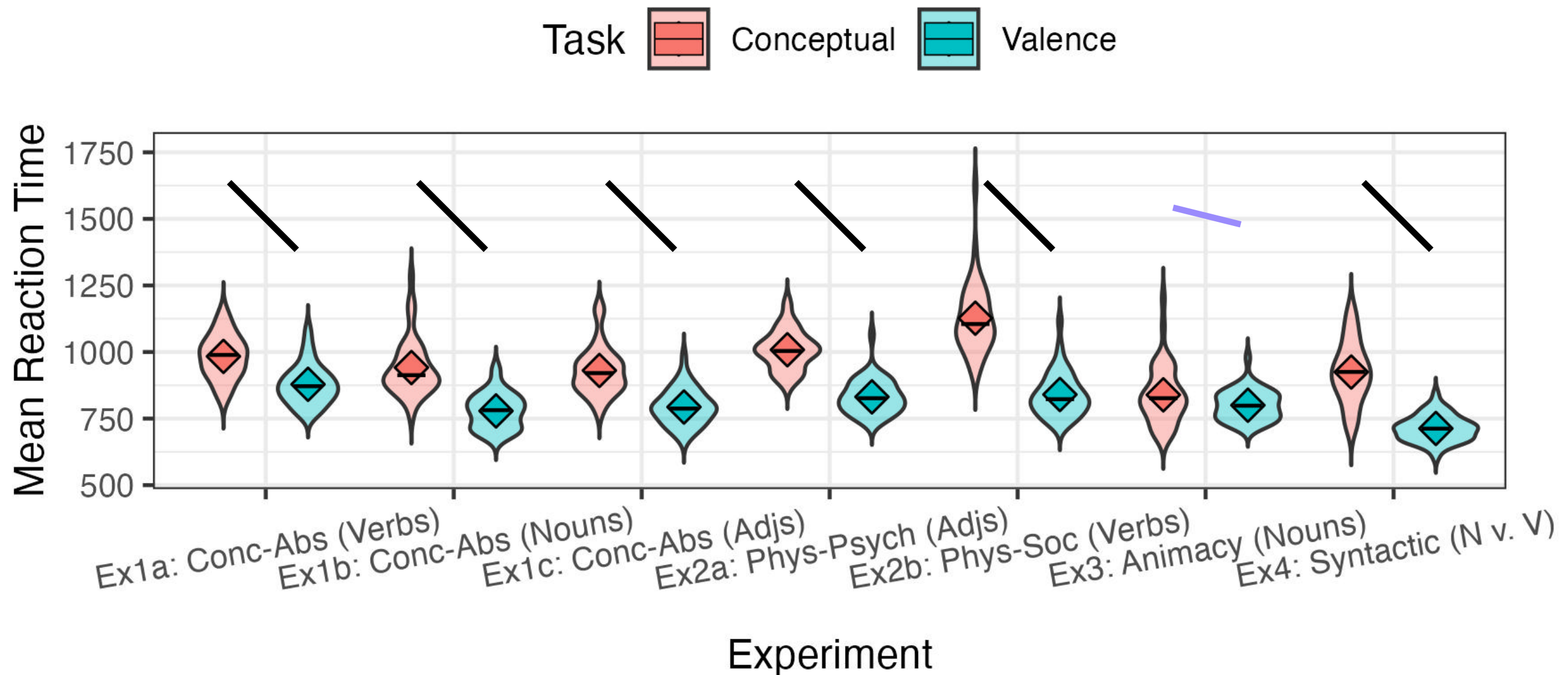
Model:

$\text{LogRT} \sim \text{Task} + (1 + \text{Task} \mid \text{ID}) + (1 + \text{Task} \mid \text{Word})$

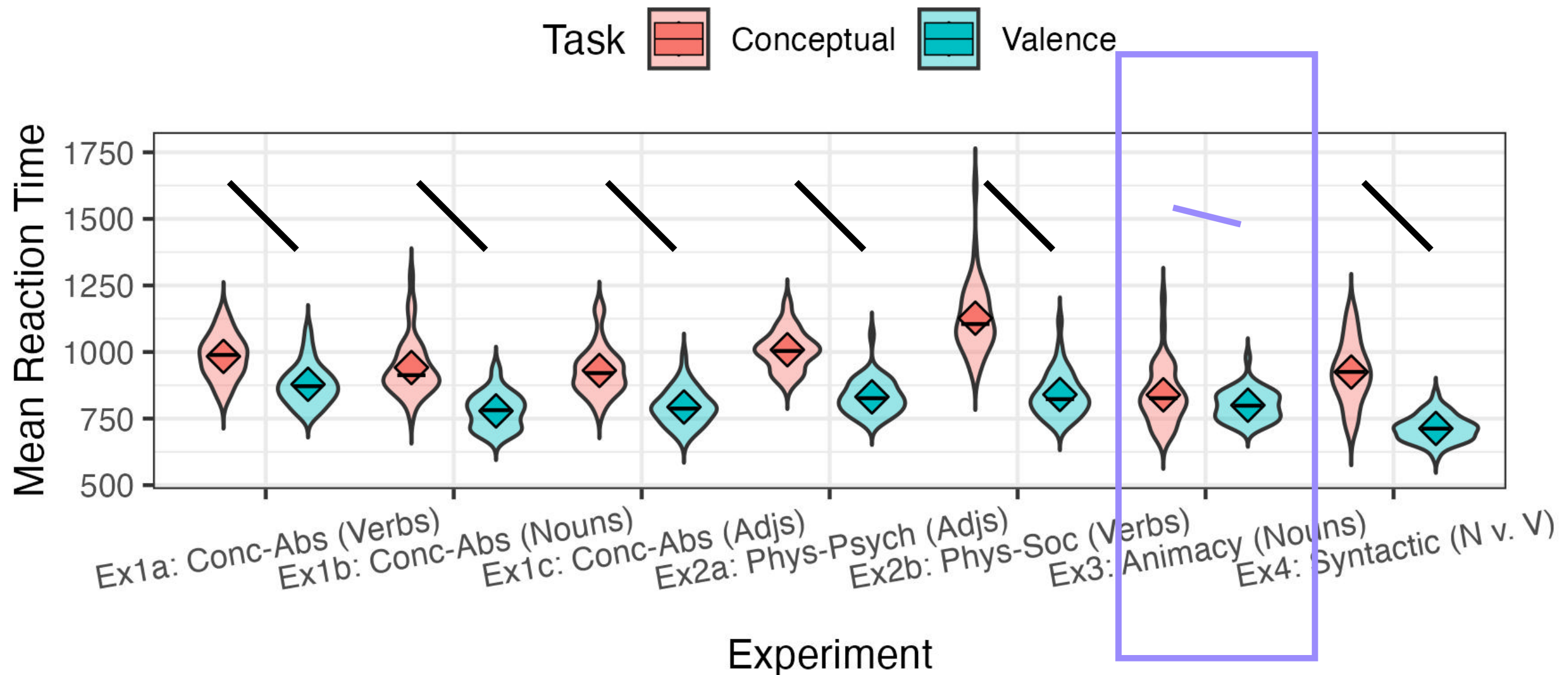
Results: Categorization



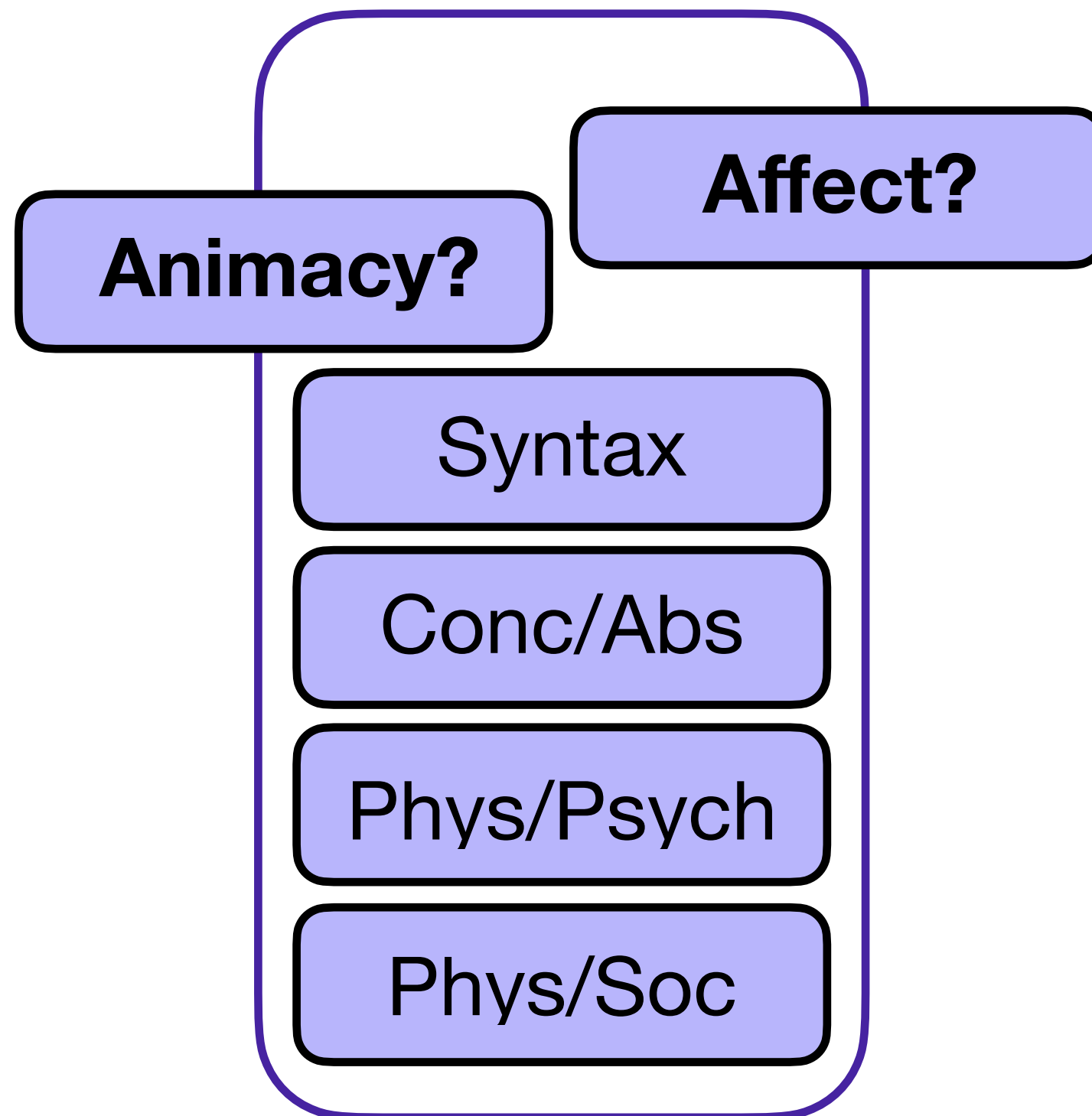
Results: Categorization



Results: Categorization

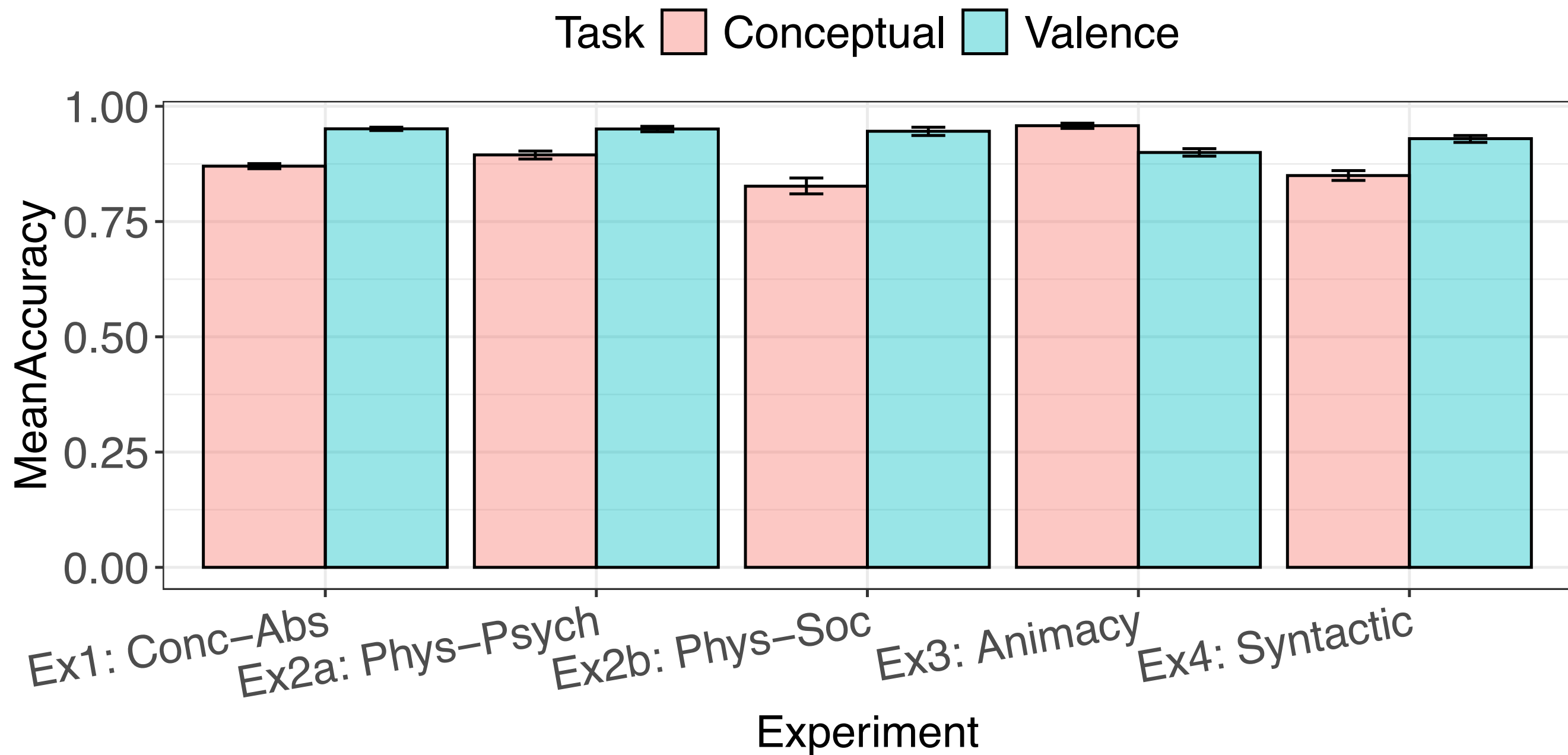


Affect ? Animacy

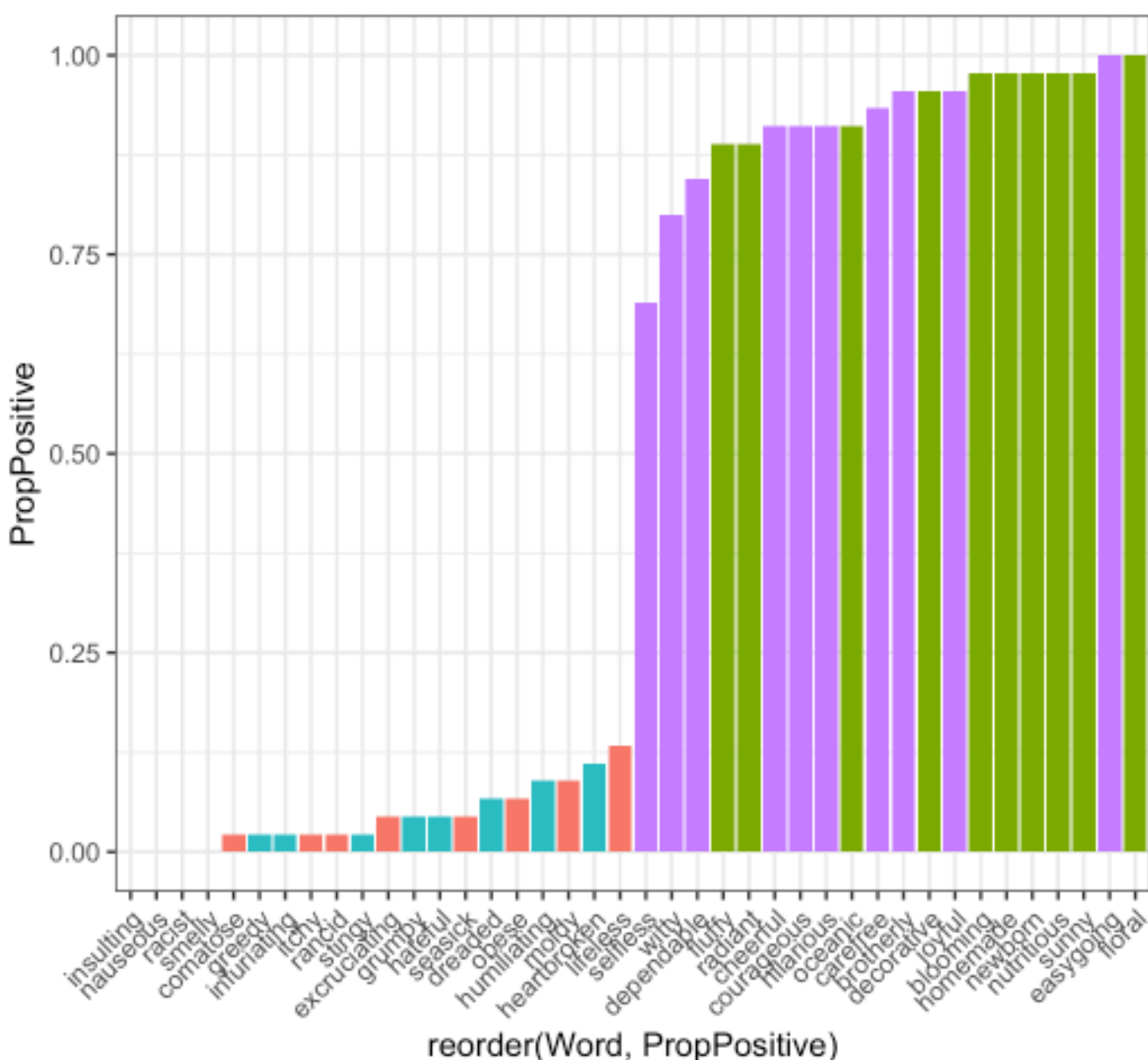
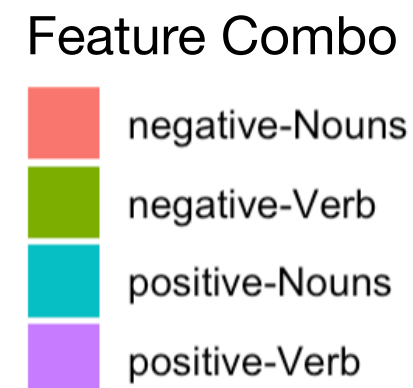


**Not just uncertainty about
conceptual meaning**

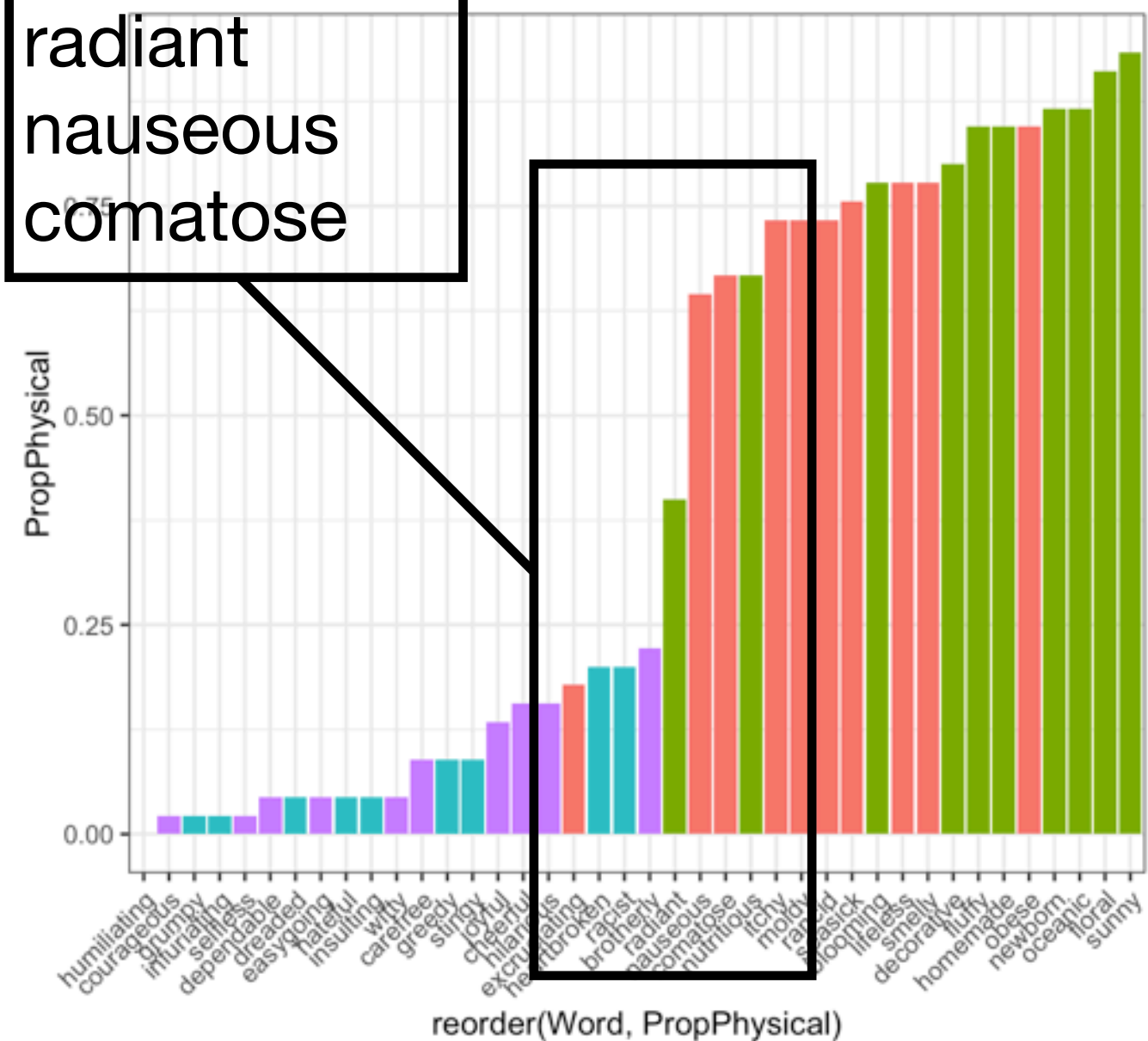
Total Accuracy



Responses Exp2b

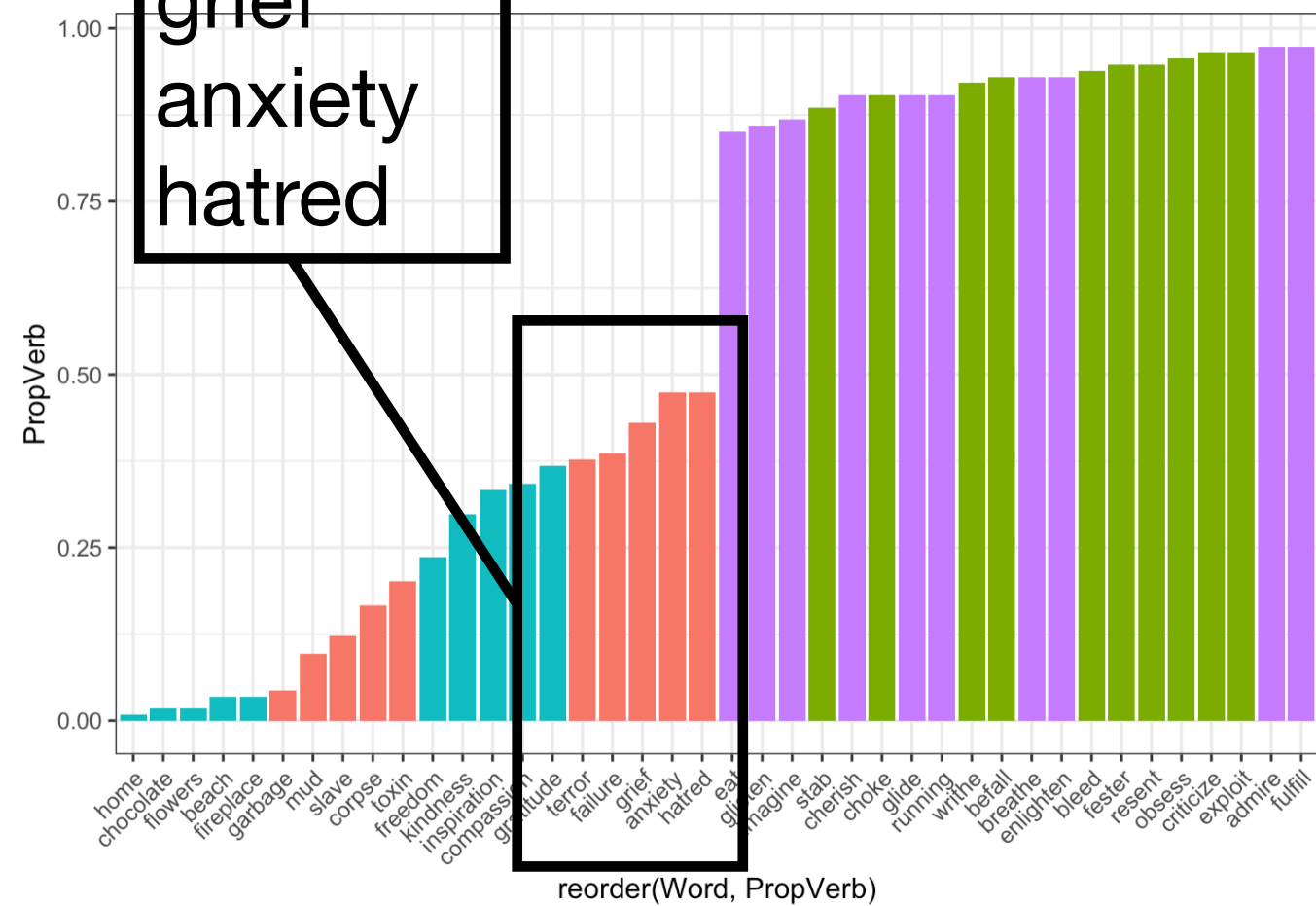
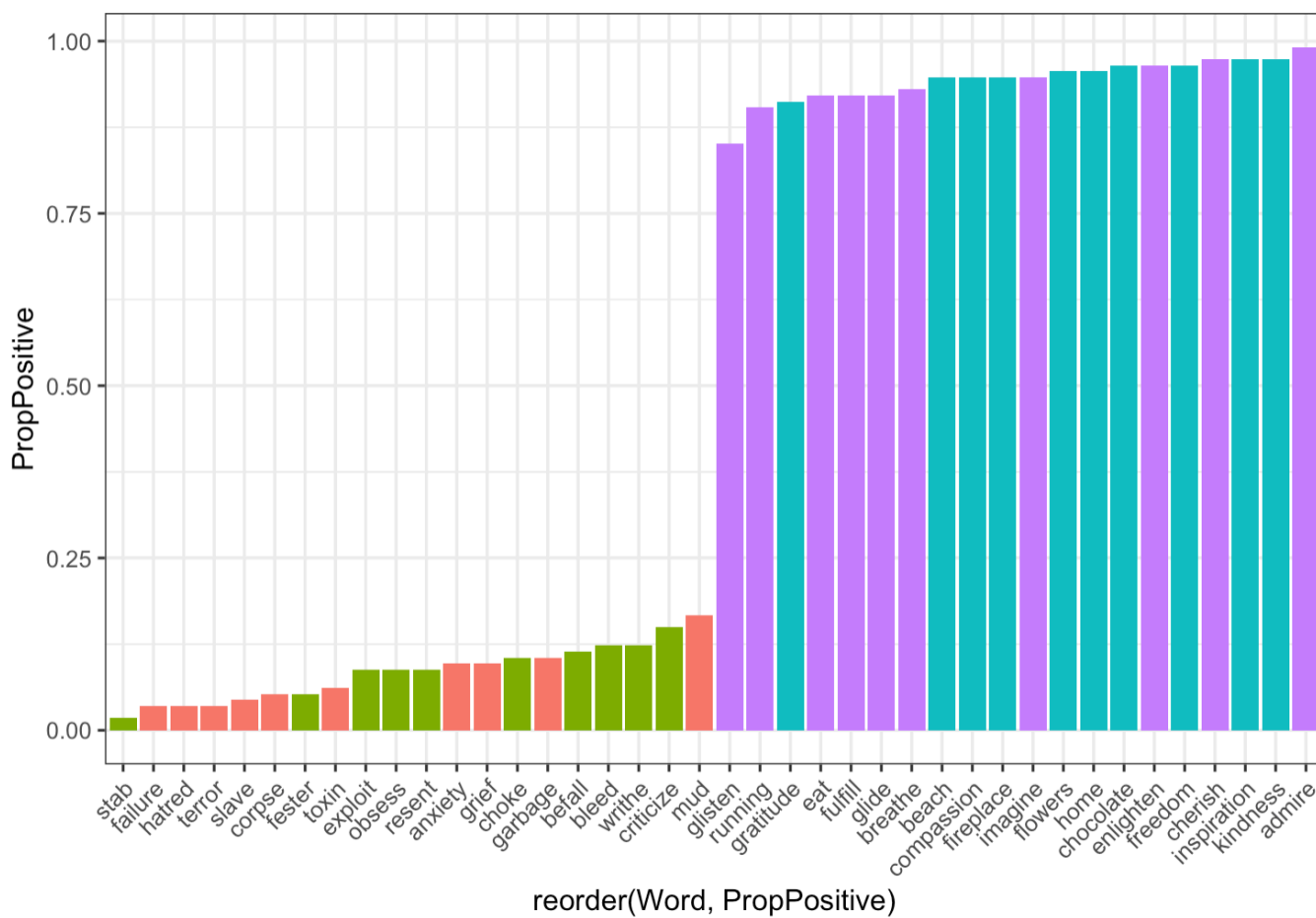
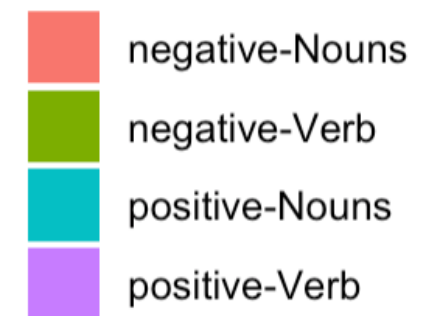


excruciating
radiant
nauseous
comatose



Responses Exp4

Feature Combo

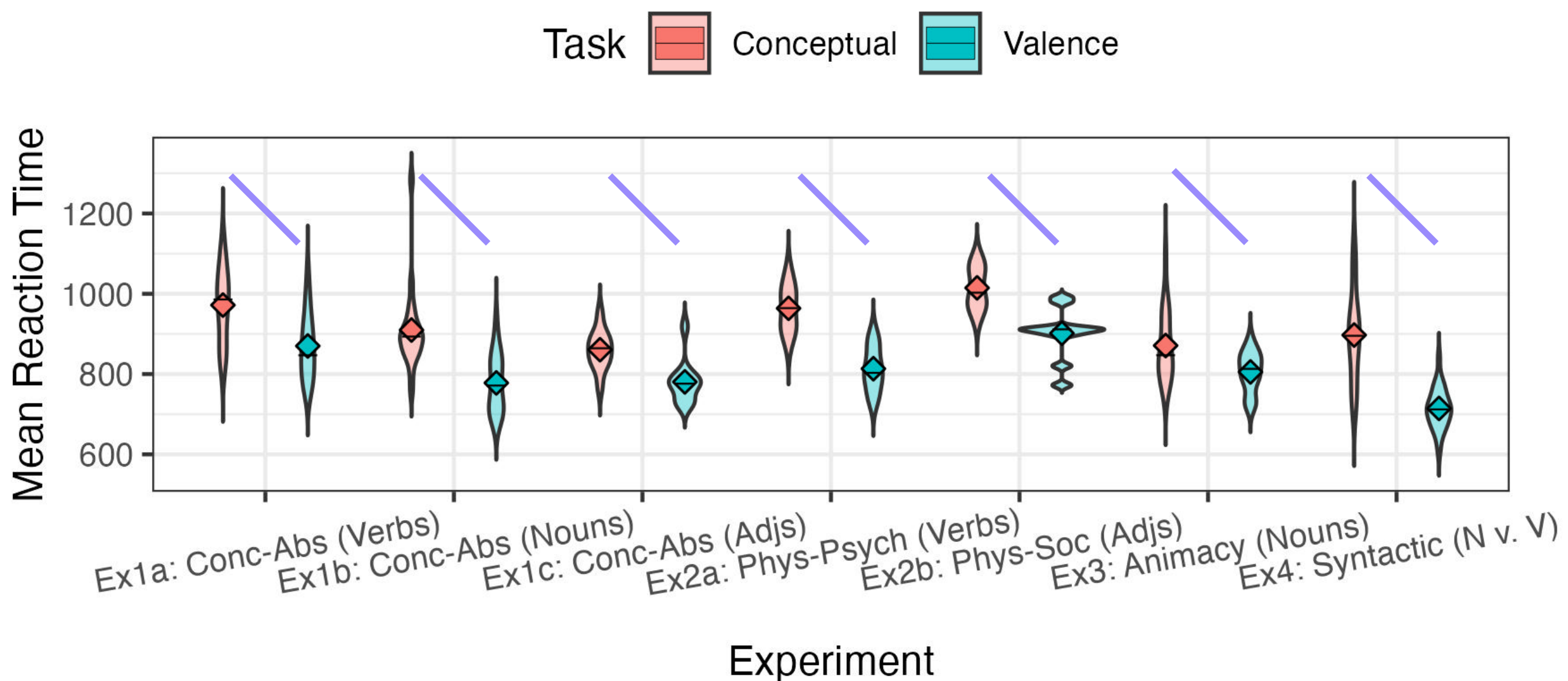


Analysis: Accuracy Control

For each study (n=7), take the 10 words where Accuracy on **Conceptual** task $\geq$ Accuracy on **Valence**

$$\text{LogRT} \sim \text{Expe} * \text{Task} + (1 + \text{Task} \mid \text{ID}) + (1 + \text{Task} \mid \text{Word})$$

RT for Most Acc. Words



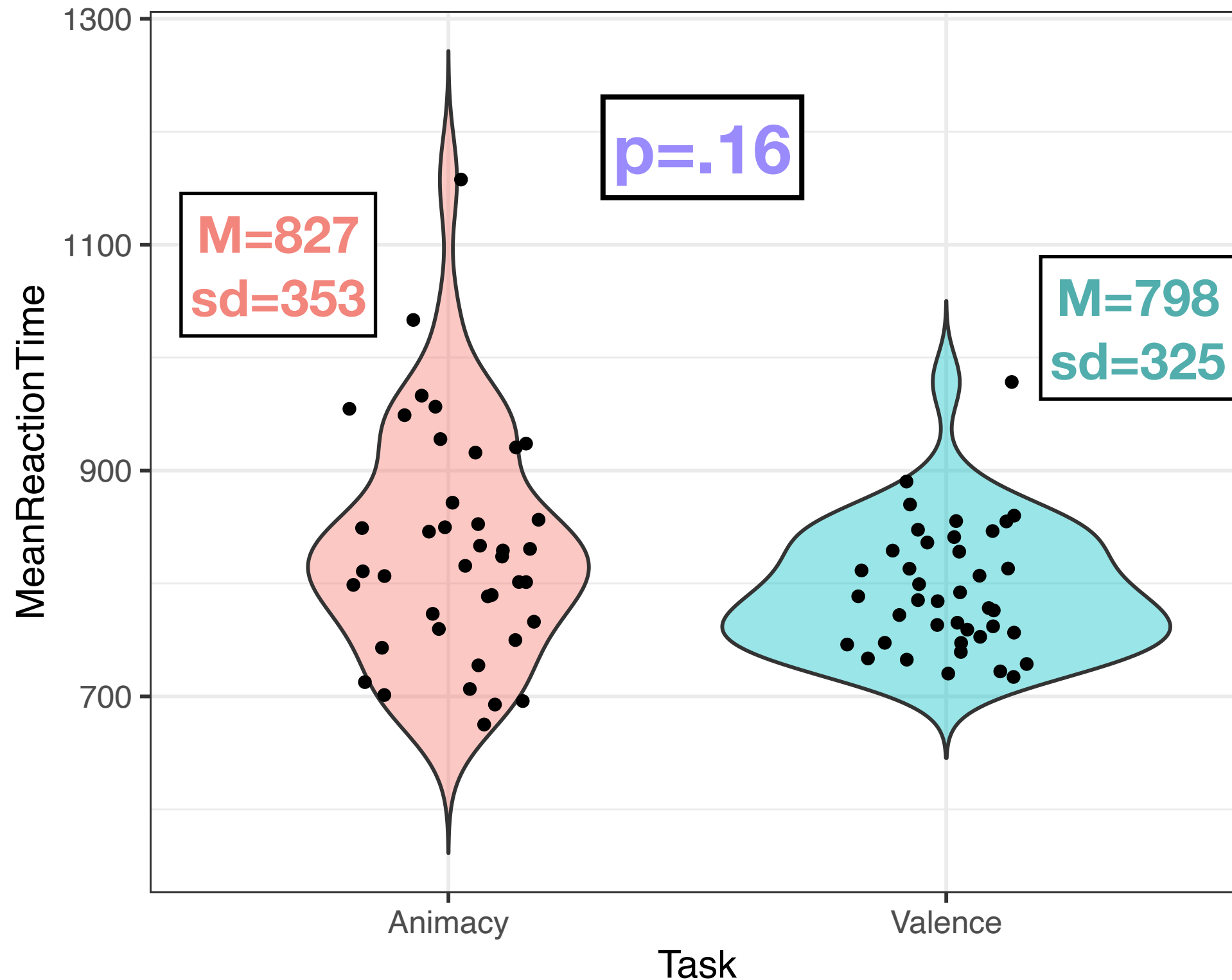
**Not just uncertainty about
conceptual meaning**

Not just Habit

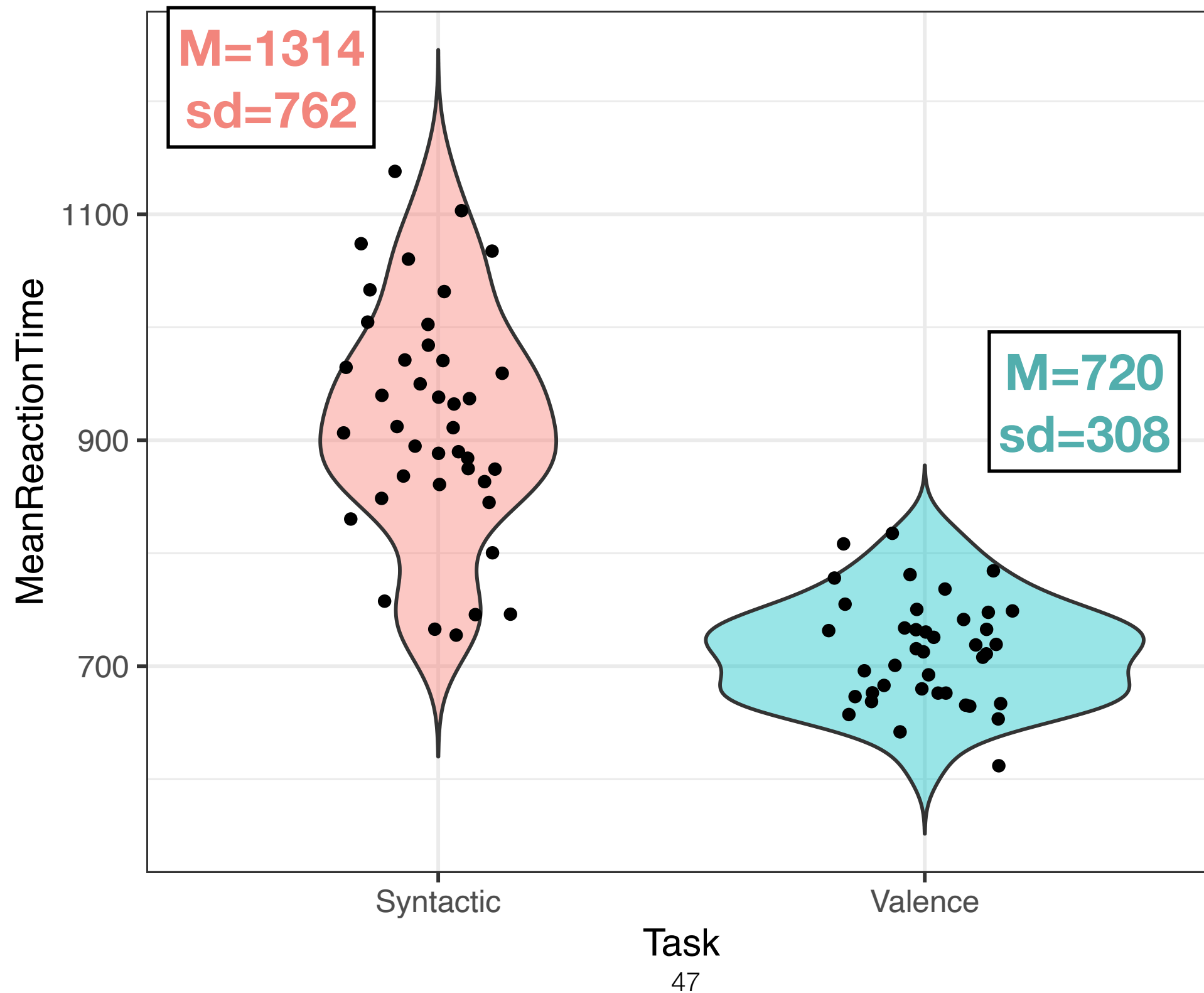
Not just uncertainty about
conceptual meaning

Not just Habit

We aren't in the habit of explicitly judging Animacy



But we do for Syntactic category when we deploy language



**Is affective info more
prominent for determining
word similarity in an implicit
task?**

progress

push
pull

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

Similarity judgements

Implicit measure of word meaning

(Osgood et al 1957; Degen 2015; De Deyne, Perfors & Navarro 2016, a.o.)

4 x 1: Feature Match

DM: slider from 0 (dissimilar) to 100 (similar)

MaxMismatch, DissimilarControl (Antonyms)

MaxMatch, SimilarControl (Synonyms)

**Matching
ValenceOnly**

vs.

**Matching
ConceptualOnly**

progress

(**con** , **val**)

push

(**con** , **val**)

pull

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

progress

(**abs** , pos)

love

(**abs** , neg)

hate

**Matching
ConceptualOnly**

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

progress

(abs , pos)

love

(con , pos)

hug

**Matching
ValenceOnly**

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

Stimuli

Words from categorization experiments (7 lists)

All possible combinations, within-list

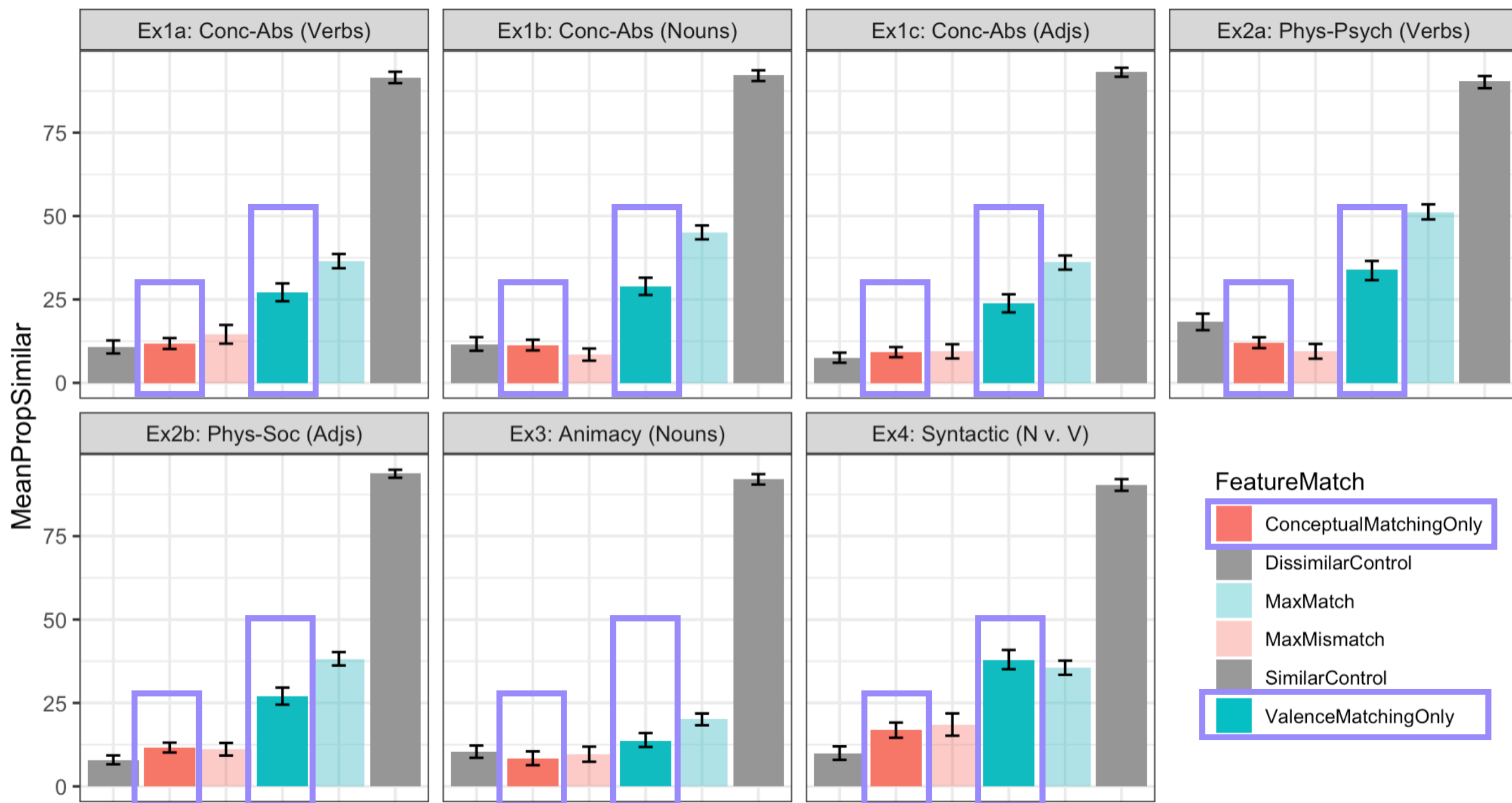
- Unordered
- Same-word pairs removed

60 items

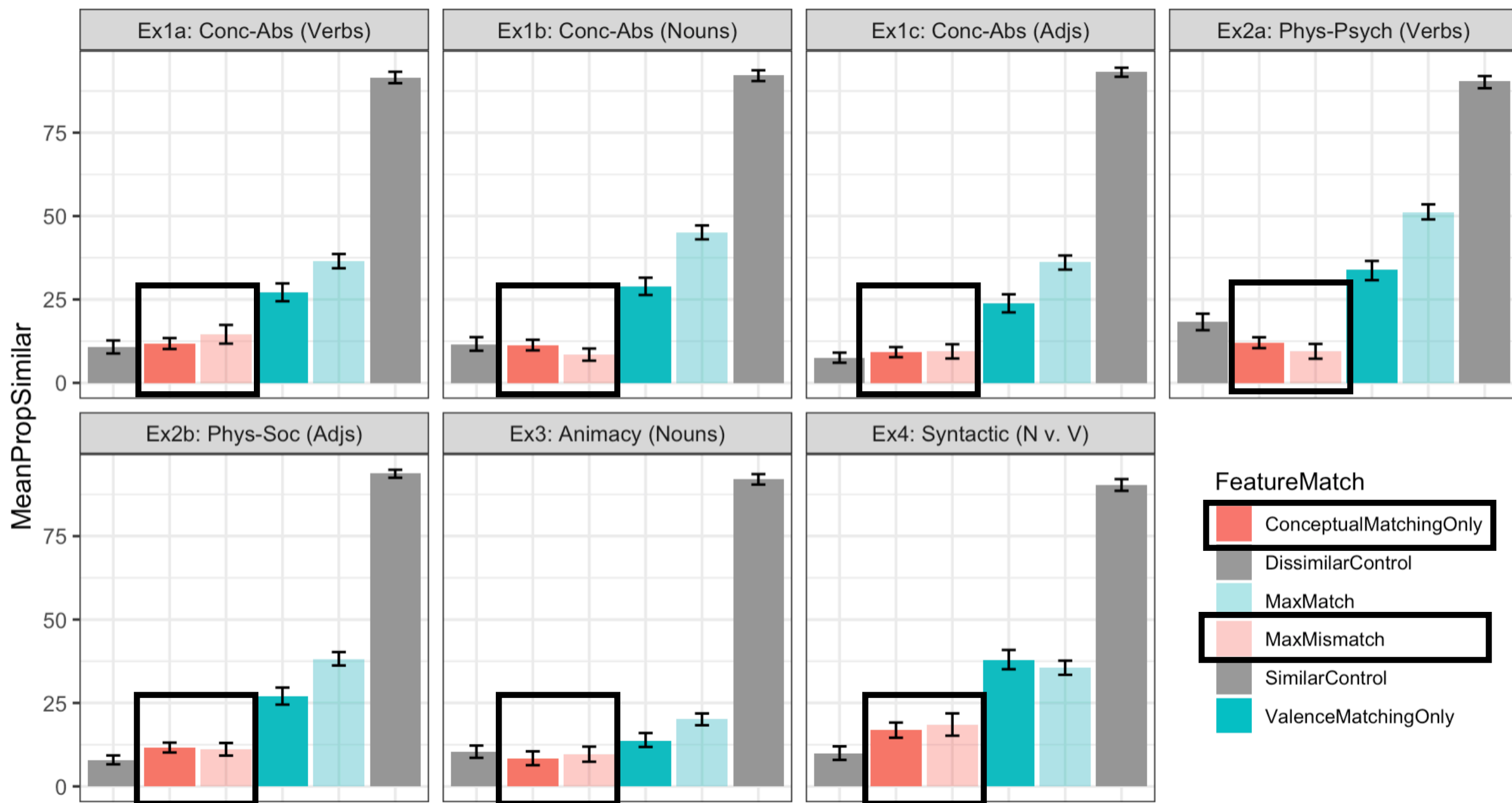
- 10 randomly sampled from the 4 FeatureMatch Conditions
- +10 Similar Controls (Synonyms),
- +10 Dissimilar controls (Antonyms)

Rating $\sim$ FeatureMatch + (1 + FeatureMatch | ID) + (1 | Word)

Results (n=350)



Results (n=350)



Support for AFH in isolated words

Valence faster than most Conceptual categories except Animacy

Persistent effect: found in majority of words tested

Valence, not Conceptual, features determine word meaning (similarity)

Conclusion

Affective information is **prominent** in both processing and use for isolated words

Crucial component to build word meaning

Affect

Animacy

Syntax

Conc/Abs

Phys/Psych

Phys/Soc

Thank you!

**Projet SU - Emergence "REV" OP24-126
Valence Asymmetries, ERC Advanced Grant,
grant n° 101142133**

Attendees of XPragFest 2025, EPITHETS-ST AHL 2025, and
New — Experimental — Perspectives on Valence 2025



Experiment 1a-c:

concrete / abstract

Exp 1a-1b (n=90)

Using Corpus norms for Valence (Warriner et al 2013)
and Concreteness (Brysbaert et al 2013)

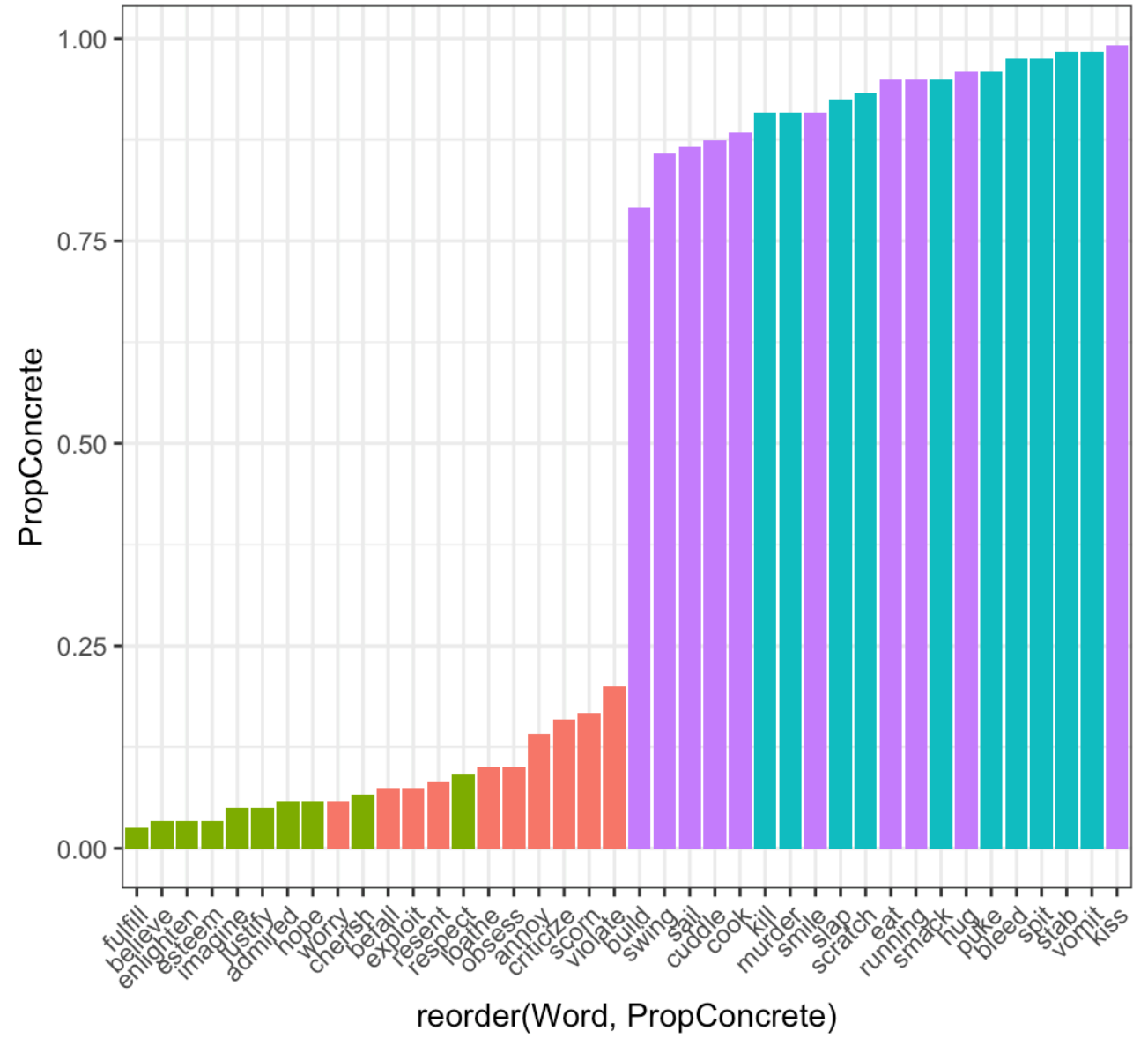
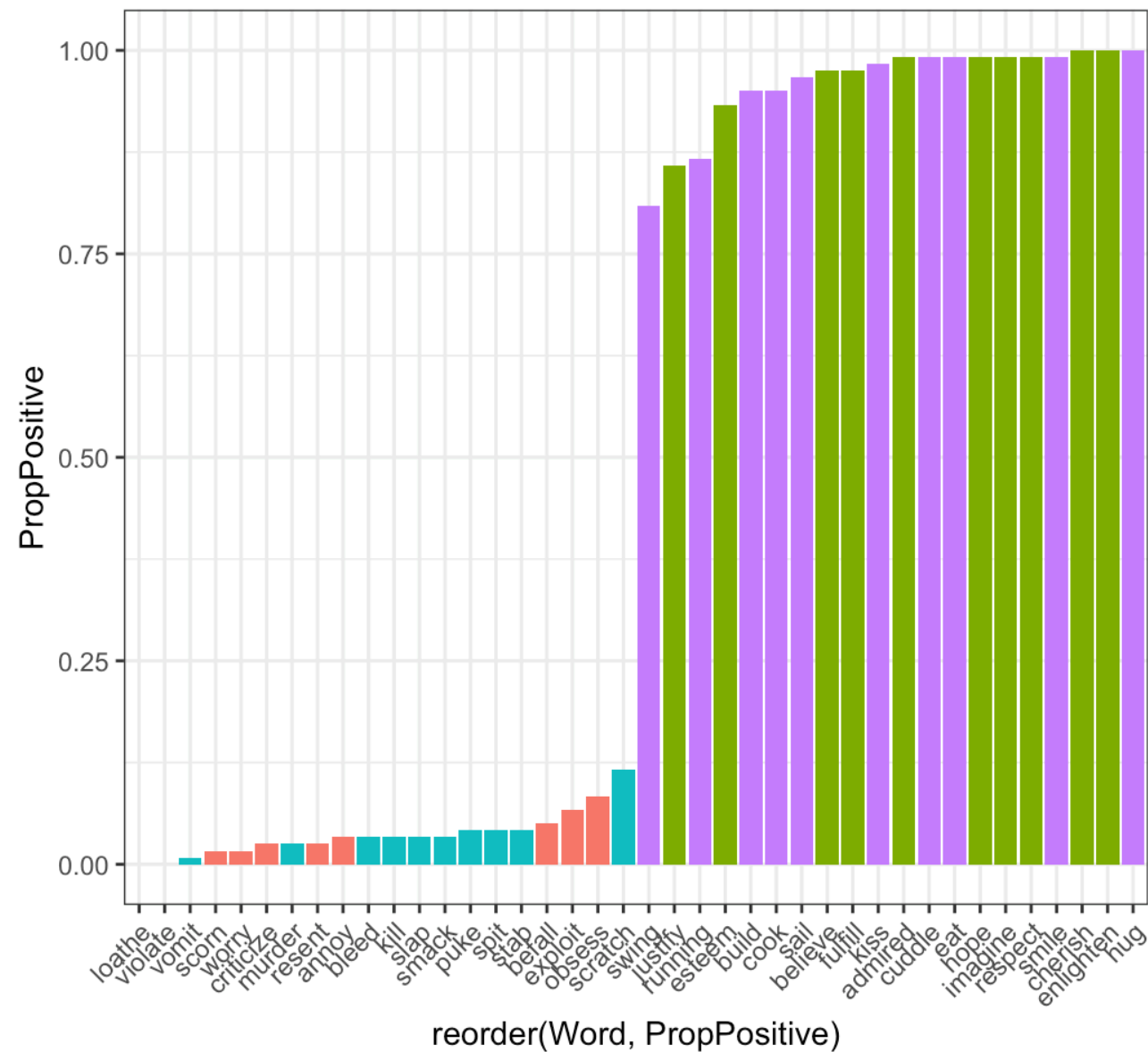
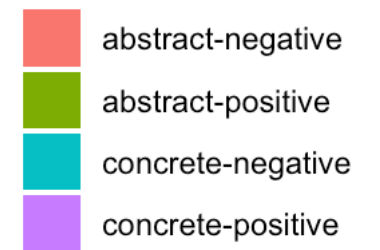
High Frequency, and within +/-2sd of mean Word Length, and overly valanced morphosyntax

40 words per, 10 from each combo of Conc+Val

- Maximizing mean rating for concrete/positive
- Minimizing mean rating for abstract/negative
- Minimizing rating SD

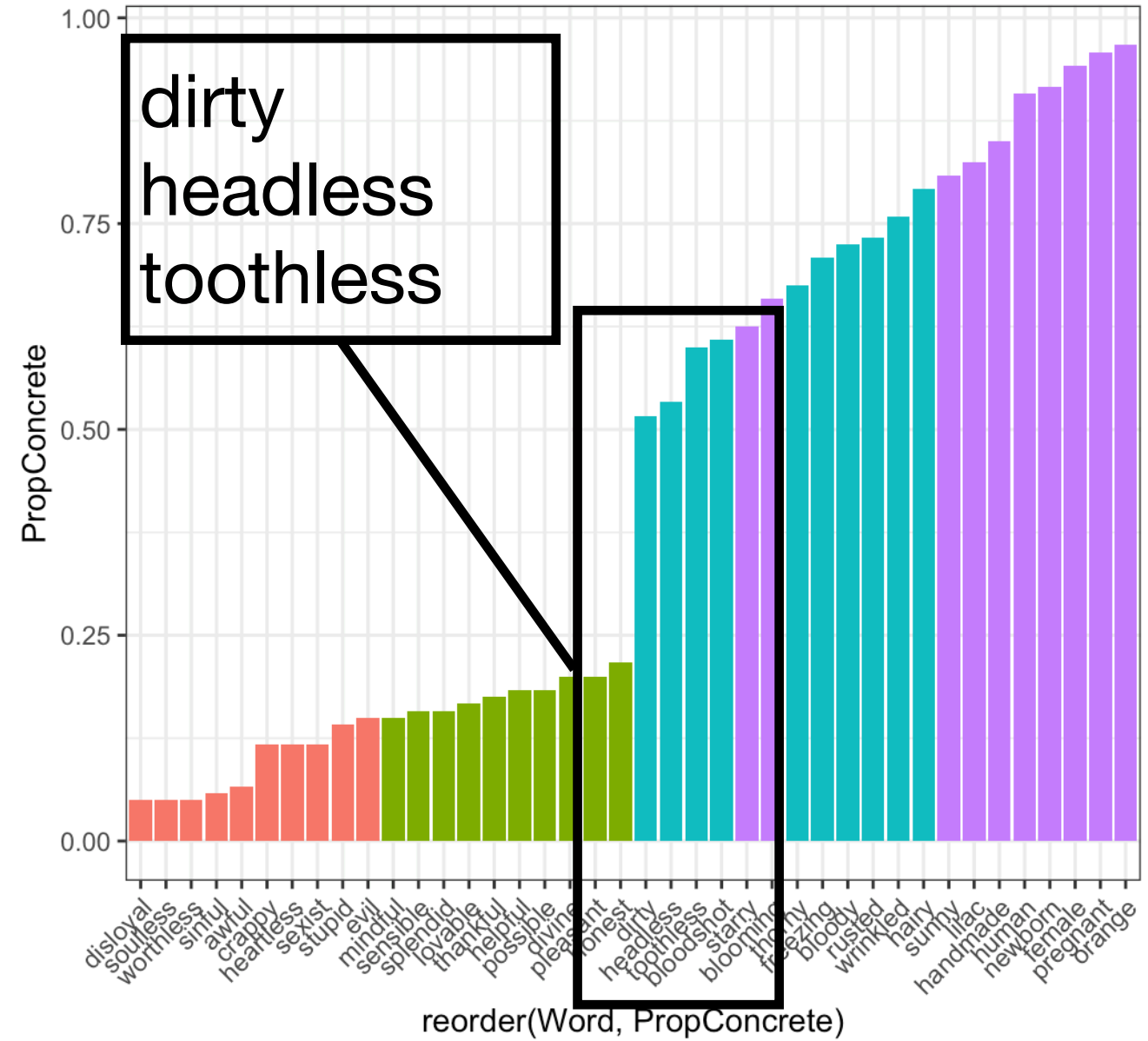
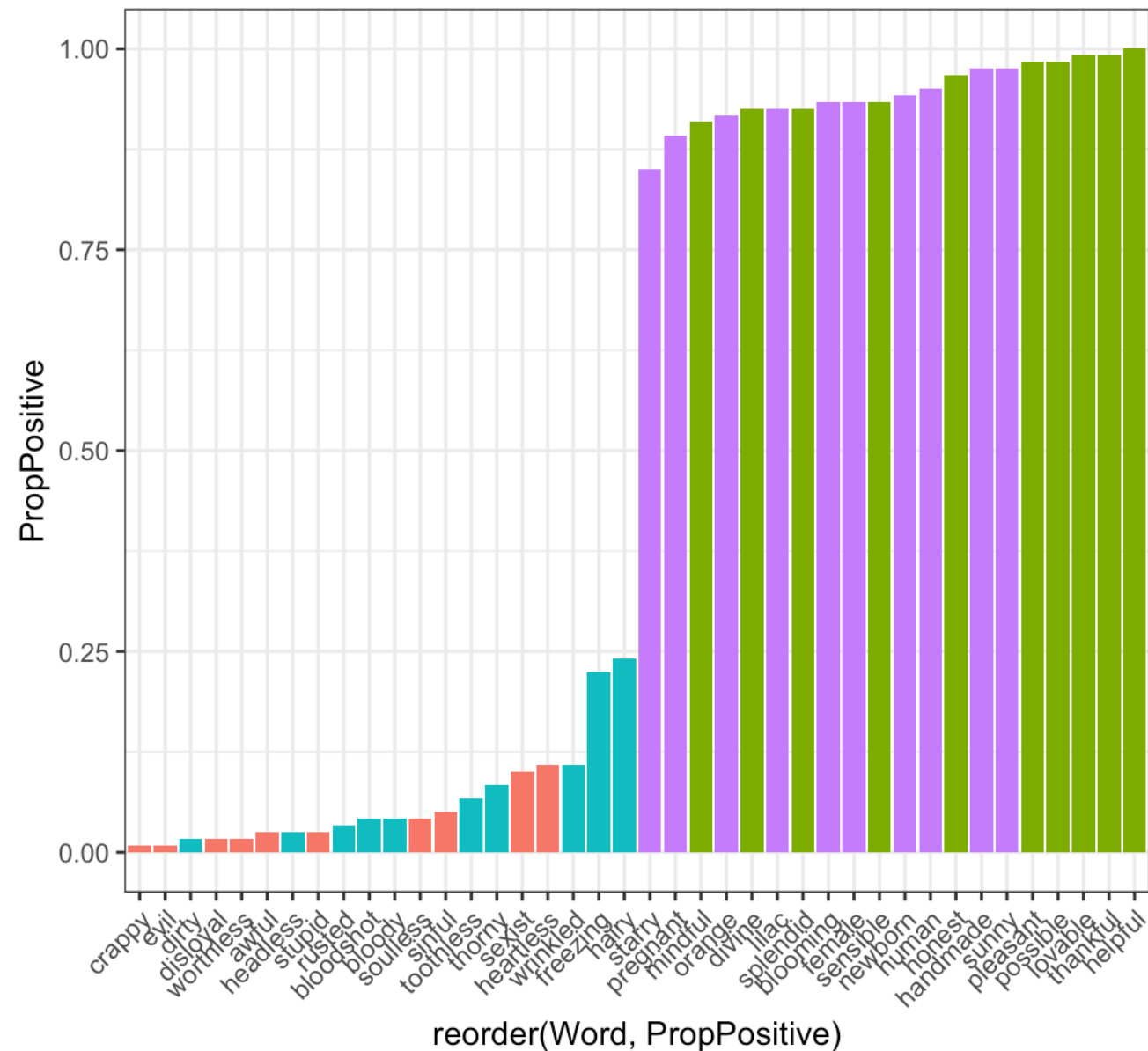
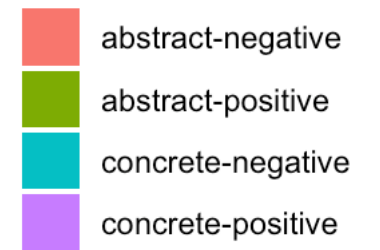
Responses - Verbs

Feature Combo

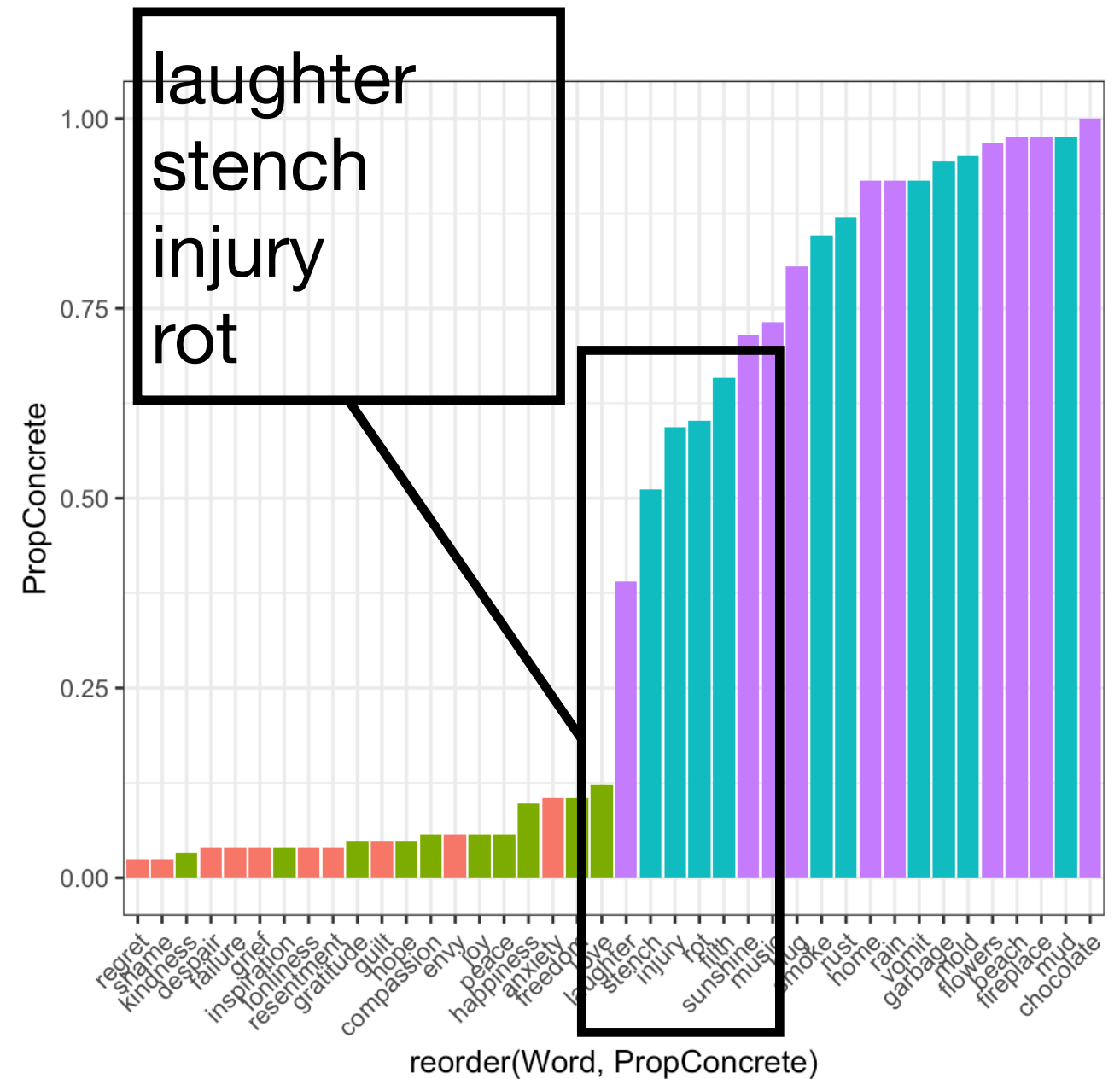
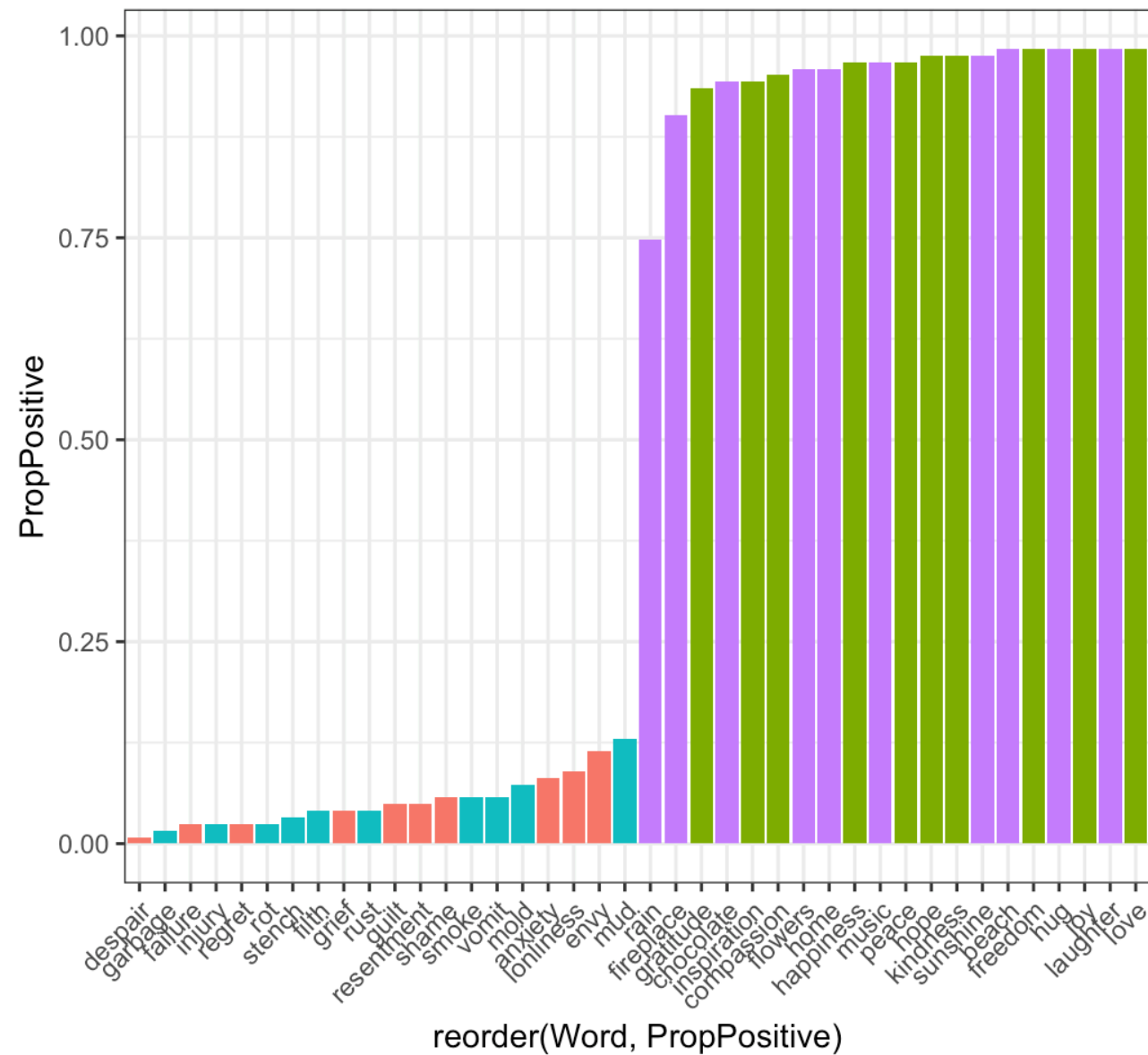
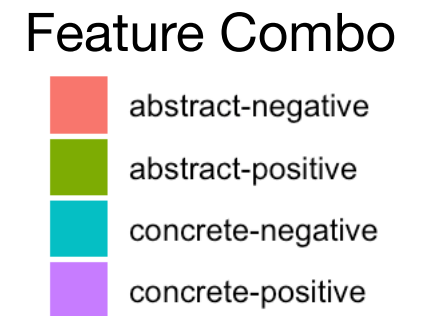


Responses - Adjs

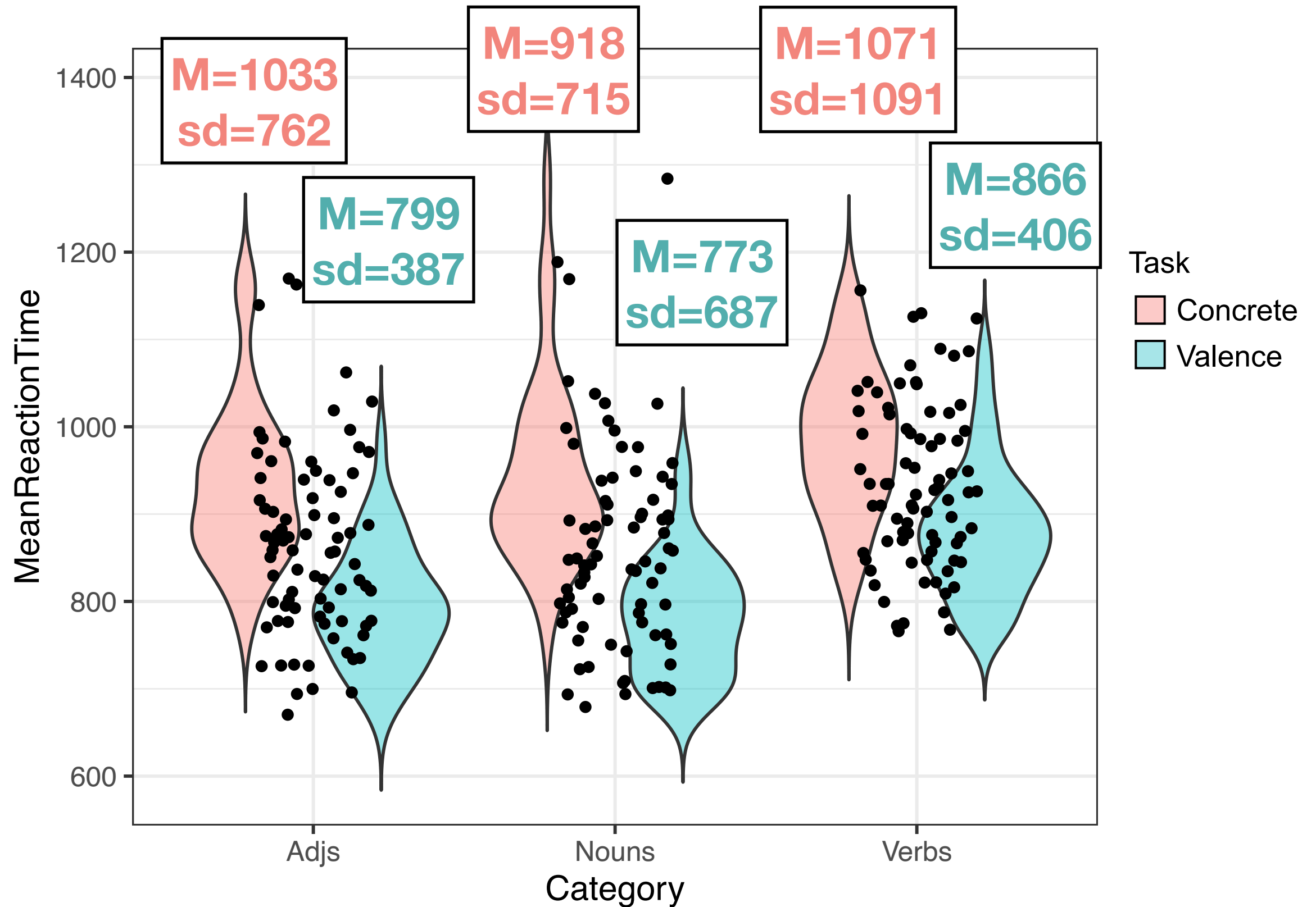
Feature Combo



Responses - Nouns



Experiment 1 Results



Adjs



Nouns

MeanReactionTime



Task



Verbs



Experiment 2a-b: physical vs psychological (verbs)/ social (adjs)

Exp 2 Stimuli

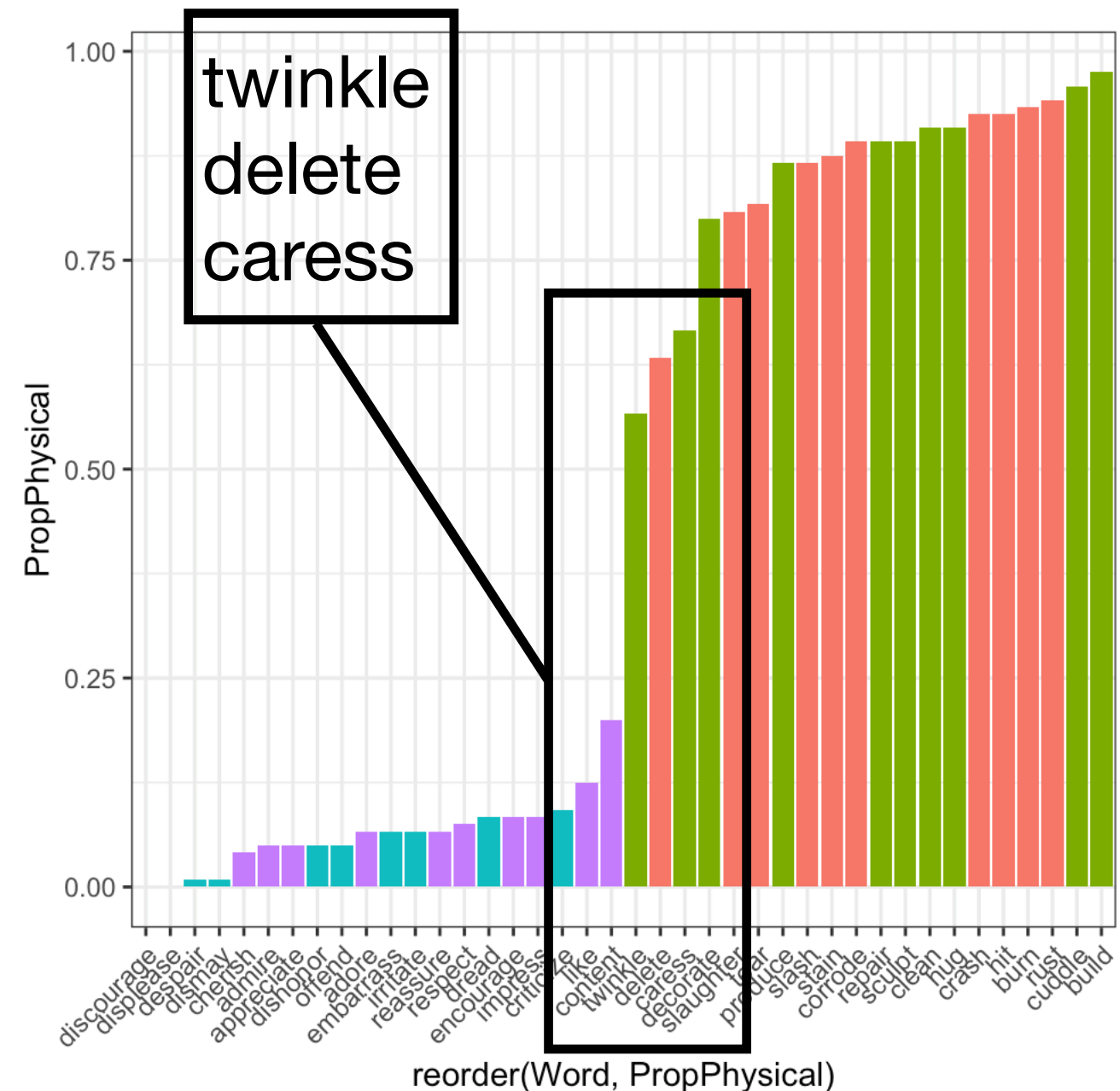
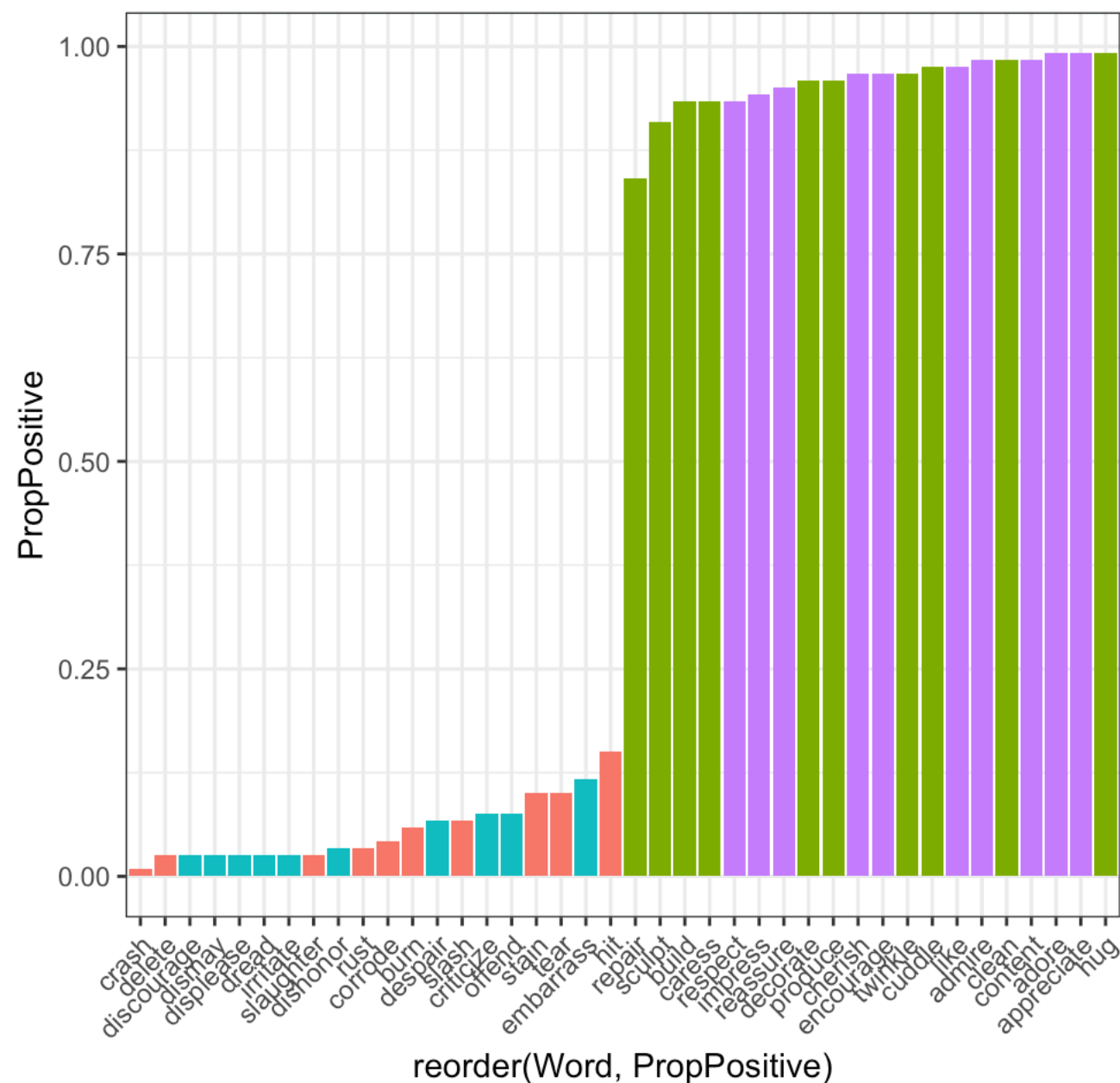
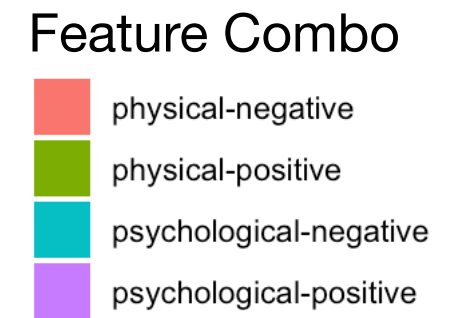
2a Verbs (n=30):

- Pre-test norming of 168 verbs from Levin (1993)
- Evaluated on Valence/Conceptual (phys-psych) 7-pt likert scale
- 40 Verbs chosen based on distance to endpoint

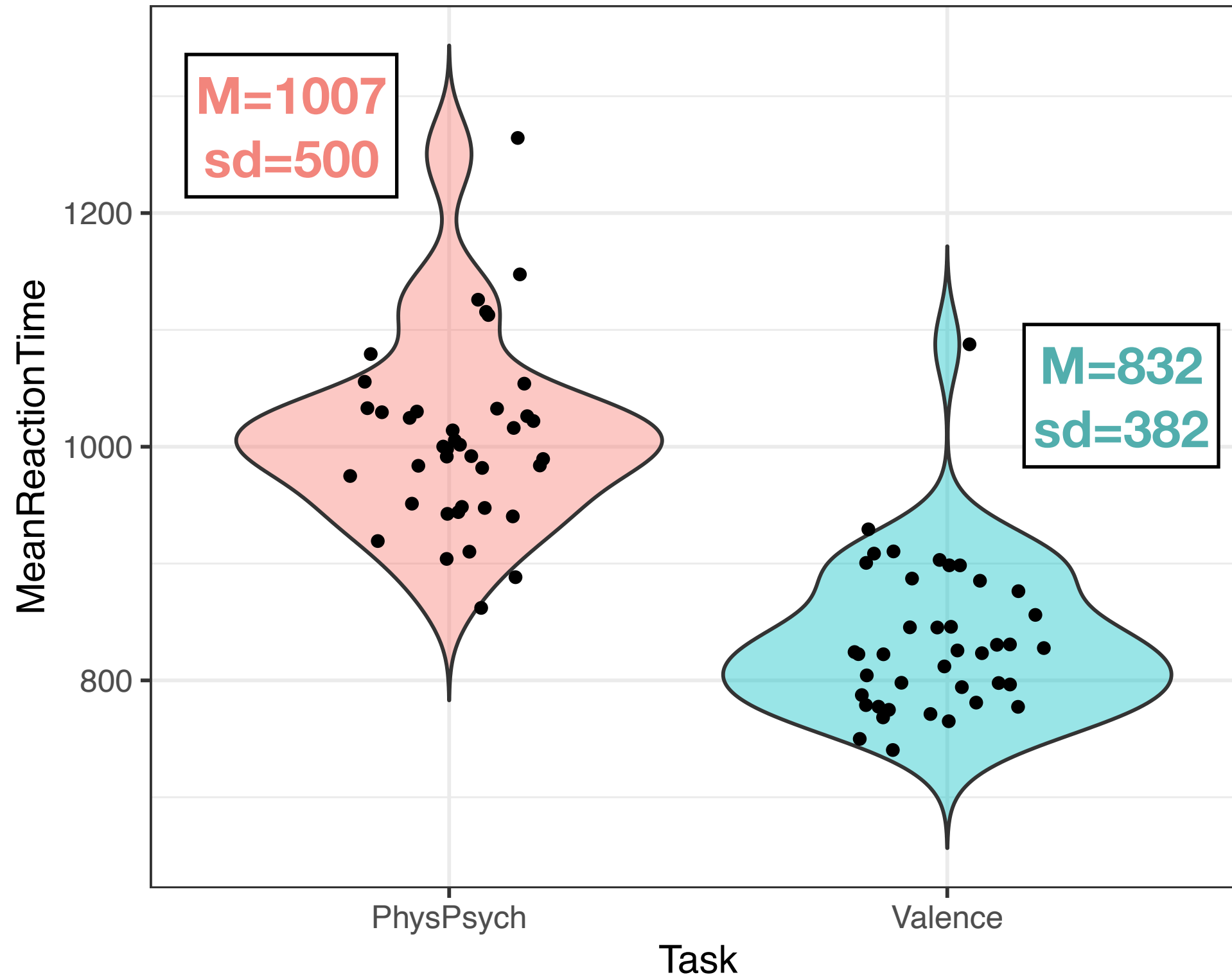
2b Adjectives (n=30):

- Pre-test norming of 289 adjectives from Warriner et al (2013)
- Evaluated Social/Physical on 7-pt Likert
- 40 Adjs chosen based on weighted average and minimized rating variance

Responses 2a



Exp 2a (Verbs) Results

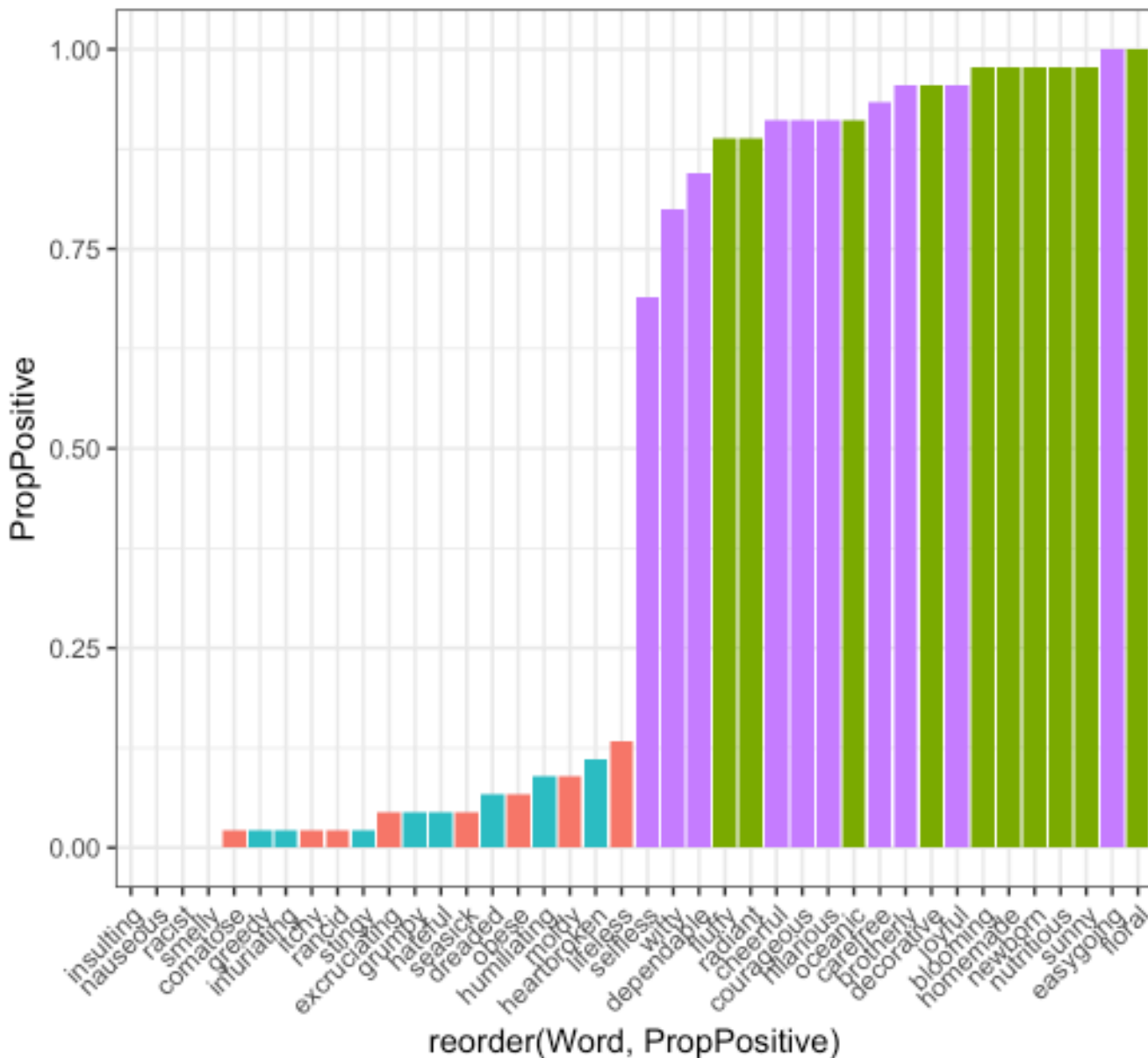


Verbs

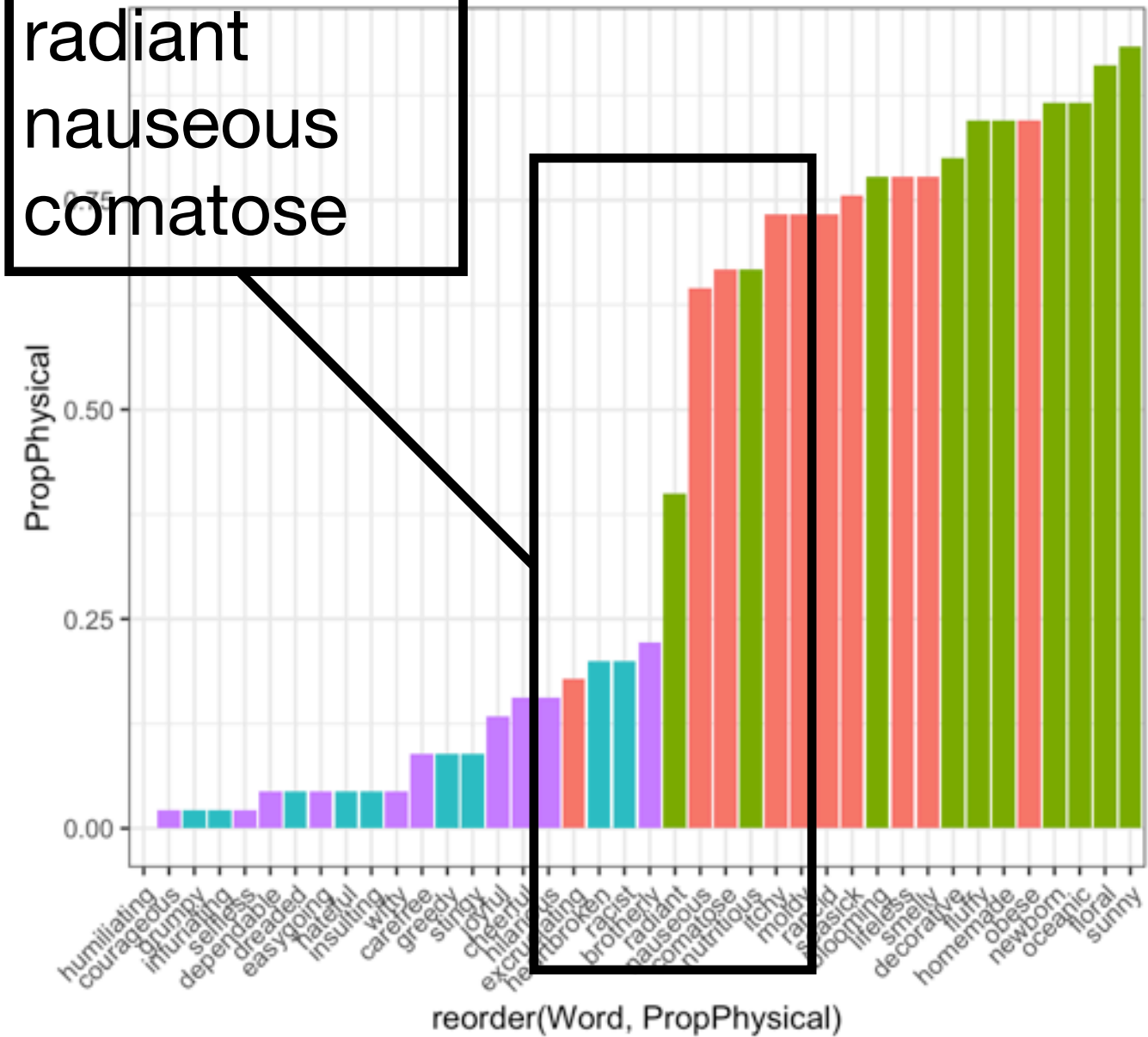


Responses 2b

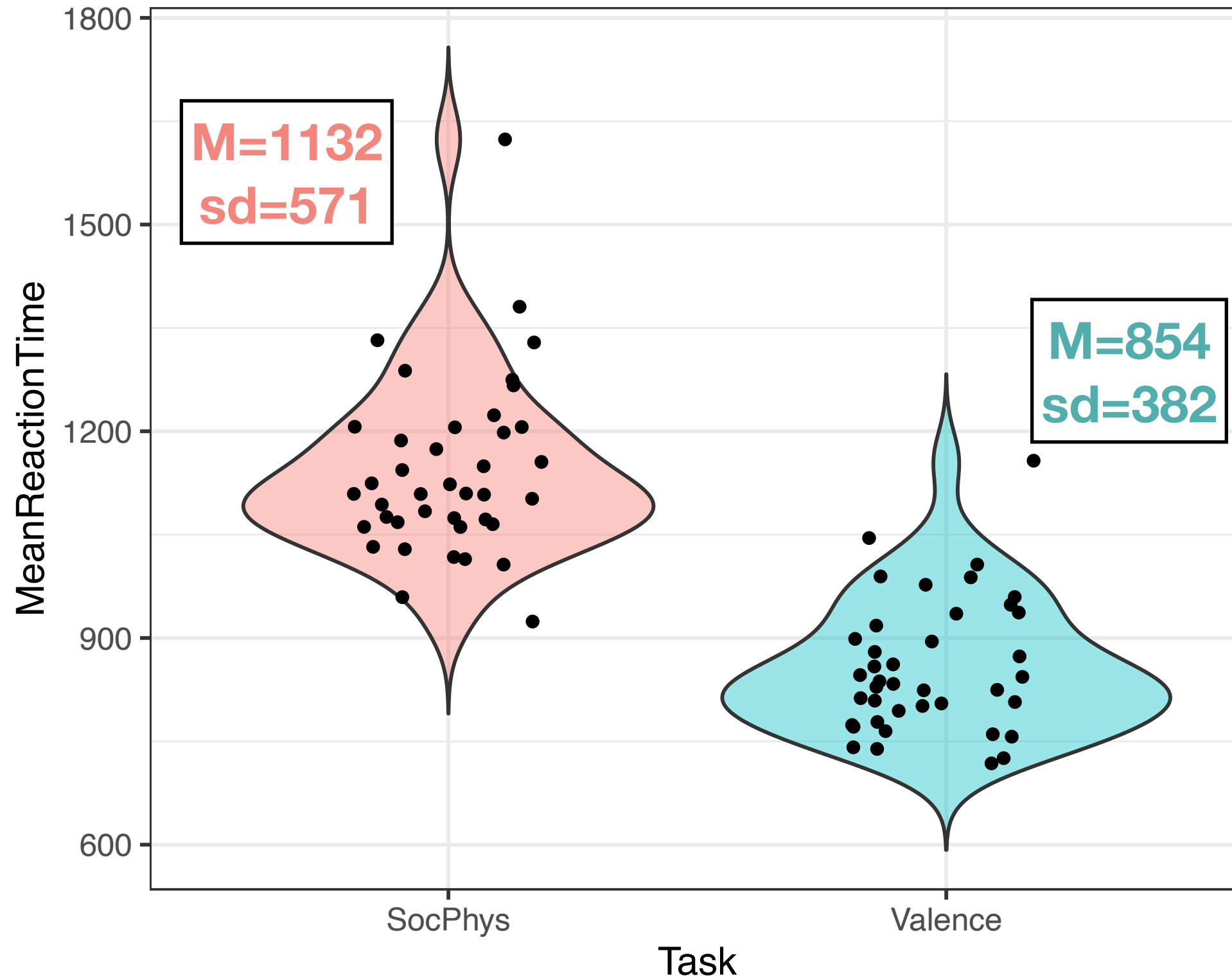
Feature Combo



excruciating
radiant
nauseous
comatose



Exp 2b (Adjs) Results



Adjs



Experiment 3:

Animacy (Nouns)

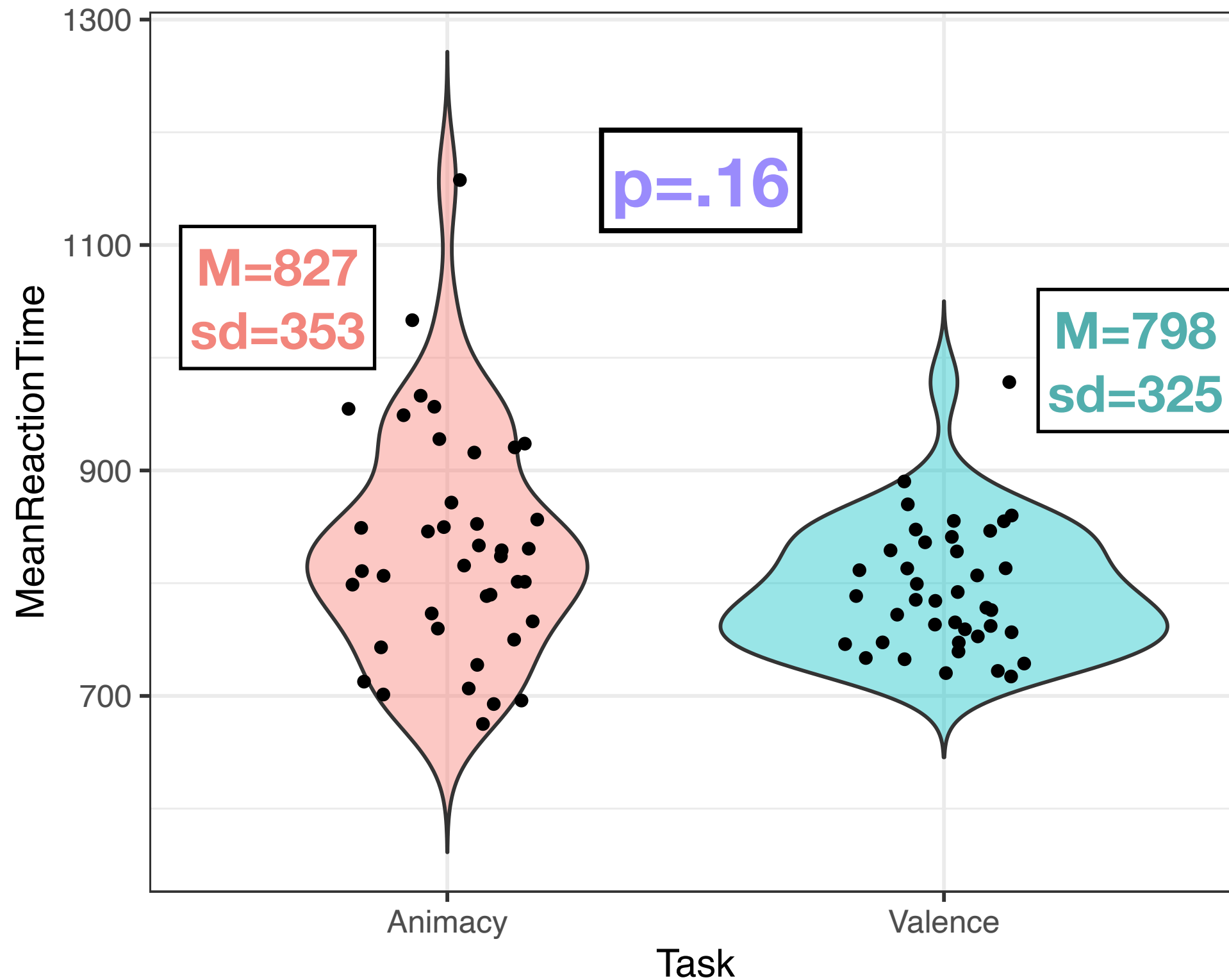
Exp 3 Stimuli

Using Corpus norms for Valence (Warriner et al 2013)

WordNet (Fellbaum 1998) words with 'living thing' or 'organism' as synset

High Frequency, and within ± 2 sd of mean Word Length

Exp 3 Results (Nouns)



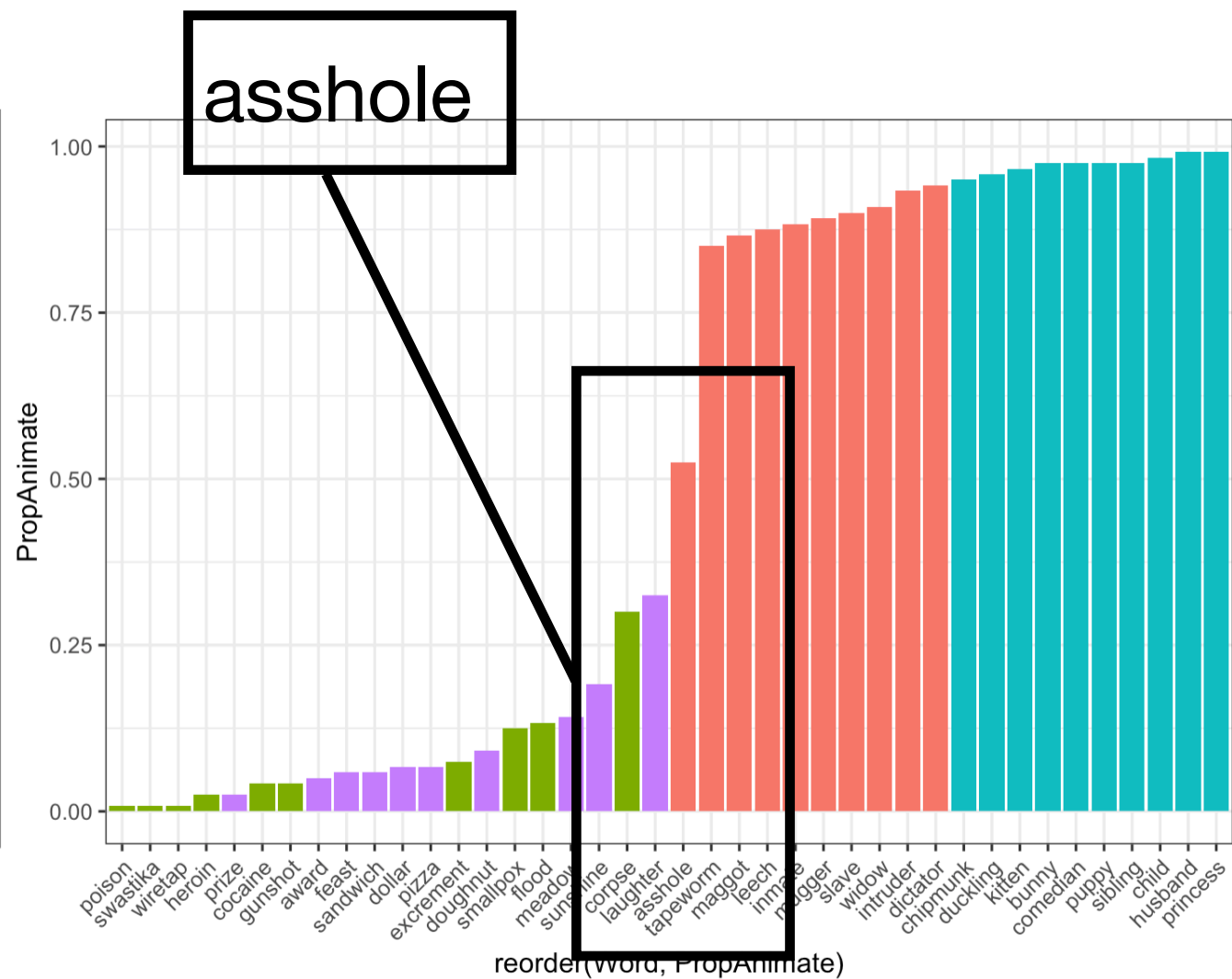
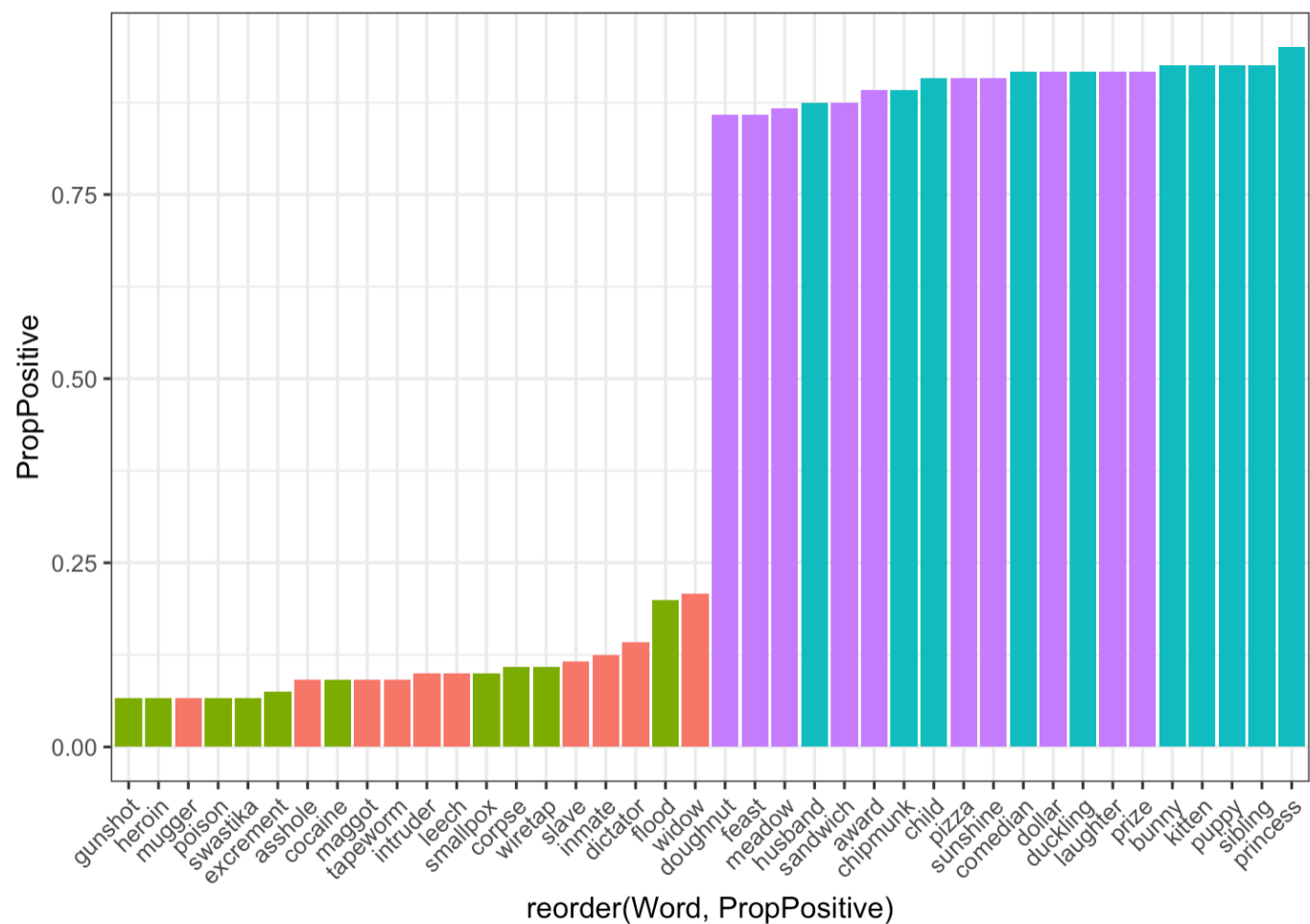
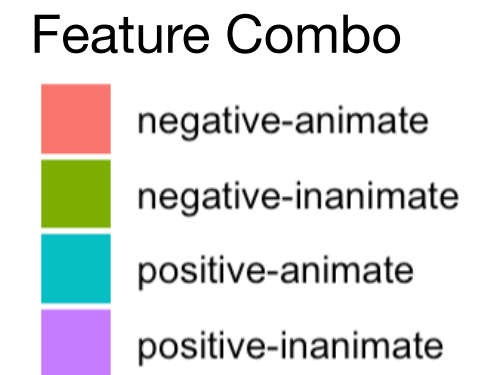
Nouns



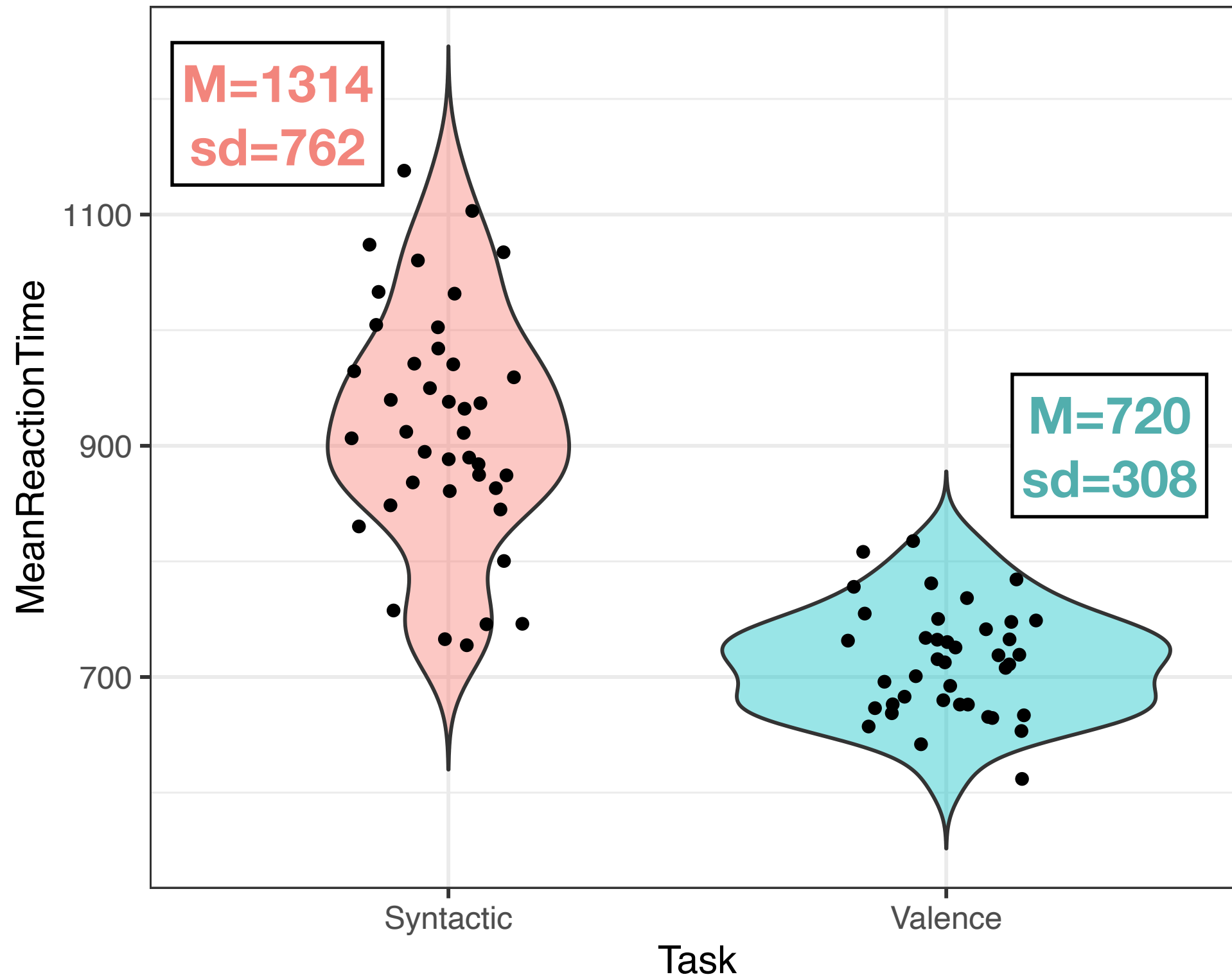
Experiment 4:

Syntax

Responses Exp3



Exp 4 Results



Verbs

Nouns

MeanReactionTime



Task



Responses Exp4

Feature Combo

